

Almost-Idempotent Stochastic Maps — internal lab-book

(scientific-exploration repo; see PRD.md / CLAUDE.md)

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1 Orientation and rigour boundary

The north star, and its status

The target of this project is the classical (stochastic) stability statement **op-classical**: there are universal constants $\eta_0, C > 0$, independent of the dimension, such that every row-stochastic Q with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta \leq \eta_0$ admits a stochastic idempotent E (**def-stochastic**) with

$$\|Q - E\|_{\infty \rightarrow \infty} \leq C\sqrt{\eta}, \quad (1)$$

the exponent $\frac{1}{2}$ being sharp.

This is OPEN. Nothing in this document proves it, and nothing in this document should be read as evidence that it is close to proved. What follows is a rigorous *partial* record: the components of the currently live route that have cleared this repository's top in-repo rigour rung.

Two pictures, one bridge

Two equivalent formulations are in play throughout. In the *stochastic picture* the object is a genuinely positive row-stochastic Q that is only approximately idempotent, with defect η measured in the $\infty \rightarrow \infty$ map norm (**def-almost-idempotent**). In the *signed picture* the object is an exact signed affine retraction $P - P\mathbf{1} = \mathbf{1}$ and $P^2 = P$ *exactly* (**def-signed-idempotent**) — whose failure of positivity is quarantined in the single scalar δ , the negative mass (**def-negative-mass**). The two are interchangeable up to universal constants, and the licence to cross between them is **lem-classical-equiv** (§2); a bound must always name the picture it lives in and cite the bridge when it crosses.

What this report contains

Four blocks, in dependency order.

1. **The bridge** (§2). **lem-classical-equiv**, the signed $\leftrightarrow$ stochastic equivalence, with explicit constants in both directions.
2. **Positive-retract hardening** (§3). **lem-prh** converts a *positive approximate retract* — a pair of positive unital maps A, M between finite commutative C^* -algebras whose composite MA is ε -close to the identity (**def-positive-approximate-retract**) — into an exact stochastic idempotent at distance $2\sqrt{2\varepsilon}$. It is the step that turns approximate retraction data into the exact object (1) asks for, at the sharp square-root rate.
3. **The COMP tier** (§§4–17, interleaved). This tier builds, over an *extended* ε - C^* -algebra (**def-extended-epsilon-cstar-algebra**), the approximate corner calculus of δ -projections: the compressed corners $\mathcal{S}_{P,Q}$, the compression maps $\text{Co}_{P,Q}$ and the compressed product (**def-compressed-corner**). Its content is that all of this is *matrix-uniform*: every estimate holds at every amplification level n with one constant. The keystones are the amplification identity $1_{M_n} \otimes \text{Co}_{P,Q} = \text{Co}_{I_n \otimes P, I_n \otimes Q}$, the uniform rectangular-product estimate, and the statement that a nonvanishing corner $\mathcal{S}_P$ is itself an *extended* $O(e)$ - C^* -algebra — the object the next tier needs.
4. **The H-CB tier** (§§11–22, interleaved). Over a closed H-CB datum (**def-hcb-datum**) this tier establishes the amplified estimates for the map $\text{Ha}_{P,R}^Q$ of **def-ha-map**: the column-Hilbert comparison, the variational identity that defines it, its column action, exact adjointness, its multiplicative defect, and the diagonal unit and upper-norm estimates — all uniform in n .

The honesty boundary

Three rules govern what may appear here.

Only af-validated results are stated as lemmas. Every numbered lemma below is a registry result whose adversarial proof tree has root node **validated** with clean taint, produced by fresh **codex** provers and *separate* fresh per-node verifiers; the orchestrator never judged a proof. Each shard records the node count and names its workspace. Nothing else in the registry — the inherited **proved-mod-audit** transcriptions, the **conjecture** rows, the **numerical** evidence, the **heuristic** arguments — is restated here as a theorem; those keep their tags in **argument/** and are surveyed, without promotion, in §23.

Every lemma is exactly as strong as its registry contract. Each statement is a typeset transcription of the one-line **contract** of its **argument/** shard, reproduced byte-verbatim beneath it. Where a contract is deliberately conditional or one-sided — the two-sided compressed-unit bound only for *nonvanishing* projections, the level-one hypotheses of the inverse tier — that shape is load-bearing and is preserved. In particular a level-one conclusion is never glossed as a matrix-uniform one.

No definition is restated. Terms resolve to their canonical **definitions/** shards, cited by **def-**identifier. Where a proof tree consumed a byte-matched auxiliary registered inside its own **af** workspace — the ε - C^* norm axioms, the compressed-product display, the one-dimensional nonvanishing bridge — it is named as such, and identified as a workspace external rather than a registry definition.

What is deliberately absent

No claim, direct or implied, that **op-classical** follows. The H-CB tier feeds the parent contract **conj-hcb**, which is **open** (**proved-mod-audit**, **af: none**); beyond it lie the EXT tier and a quarantined assembly interface. The route is a route, not a proof. Also absent: any numerical statement (those live in **runs/**, and are never rigorous), and the earlier signed-geometry material, which is off the live route and inventoried in §23.

2 The signed-stochastic bridge

Lemma 1 (`lem-classical-equiv`). *The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants. There is a universal $C > 0$ such that:*

1. *(stochastic $\rightarrow$ signed) if Q is row-stochastic with $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq \eta$, then $P = \theta(2Q - I)$ is a signed affine retraction (**def-signed-idempotent**) satisfying $\|P - Q\|_{\infty \rightarrow \infty} \leq C\eta$ and negative mass $\delta \leq C\eta$;*
2. *(signed $\rightarrow$ stochastic) conversely, row-normalising the positive parts p_i^+ of a signed idempotent P yields a row-stochastic Q with $\|P - Q\|_{\infty \rightarrow \infty} \leq 2\delta$ and $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq 6\delta + 4\delta^2$.*

Transcription note. The contract writes the map norm as $\|\cdot\|$ without a subscript; per `CONVENTIONS.md` (b) it is the $\infty \rightarrow \infty$ map norm $\|\cdot\|_{\infty \rightarrow \infty}$. The operator θ is carried verbatim from the contract, which does not expand it; the validated tree instantiates it as the functional calculus $\theta(X) = \frac{1}{2}(I + \text{sgn } X)$ of **def-theta-idempotent-approximation**.

Registry contract (`verbatim`).

The signed-idempotent and stochastic-idempotent formulations of classical stability are equivalent up to universal constants: Q row-stochastic with $\|Q^2 - Q\| \leq \eta$ gives $P = \theta(2Q - I)$ signed affine retraction with $\|P - Q\| \leq C\eta$ and neg mass $\delta \leq C\eta$, and conversely row-normalising p_i^+ gives Q with $\|P - Q\| \leq 2\delta$, $\|Q^2 - Q\| \leq 6\delta + 4\delta^2$.

Proof (**prose account of the af-validated tree**). *Forward direction.* Put $X = 2Q - I$. Row-stochasticity gives $X\mathbf{1} = \mathbf{1}$ and $\|X\|_{\infty \rightarrow \infty} \leq 3$, while the algebraic identity

$$X^2 - I = 4(Q^2 - Q) \tag{2}$$

converts the almost-idempotence hypothesis into $\|X^2 - I\|_{\infty \rightarrow \infty} \leq 4\eta$. Write $Y = X^2 - I$. Two properties of Y drive everything: $Y\mathbf{1} = 0$ (because $X\mathbf{1} = \mathbf{1}$ forces $X^2\mathbf{1} = \mathbf{1}$), and $\|Y\|_{\infty \rightarrow \infty} \leq 4\eta \leq \frac{1}{2}$ once $\eta \leq \frac{1}{8}$.

The smallness of Y puts the inverse square root inside its convergence radius: the binomial series

$$B = (I + Y)^{-1/2} = \sum_{m \geq 0} \binom{-\frac{1}{2}}{m} Y^m$$

converges absolutely in the map norm. Being a norm limit of polynomials in $Y = X^2 - I$, the operator B commutes with X ; since $Y\mathbf{1} = 0$ kills every positive power, $B\mathbf{1} = \mathbf{1}$; and the absolute coefficient sum gives the quantitative bound $\|B - I\|_{\infty \rightarrow \infty} \leq (1 - t)^{-1/2} - 1 \leq 2t$ at $t = \|Y\|_{\infty \rightarrow \infty} \leq \frac{1}{2}$, hence $\|B - I\|_{\infty \rightarrow \infty} \leq 8\eta$. The scalar identity $((1 + z)^{-1/2})^2(1 + z) = 1$ transfers to the absolutely convergent series in Y , so $B^2X^2 = I$; commutation then makes $S = XB$ an exact involution,

$$S^2 = XBXB = X^2B^2 = I, \quad S\mathbf{1} = XB\mathbf{1} = X\mathbf{1} = \mathbf{1},$$

with $S - X = X(B - I)$ and therefore $\|S - X\|_{\infty \rightarrow \infty} \leq 3 \cdot 8\eta = 24\eta$.

Setting $P = \theta(X) = \frac{1}{2}(I + S)$ now gives $P\mathbf{1} = \mathbf{1}$ and $P^2 = \frac{1}{4}(I + 2S + S^2) = \frac{1}{2}(I + S) = P$, so P is an exact signed idempotent in the sense of **def-signed-idempotent**. Since $Q = \frac{1}{2}(I + X)$,

the difference is $P - Q = \frac{1}{2}(S - X)$ and hence $\|P - Q\|_{\infty \rightarrow \infty} \leq 12\eta$. Finally the negative mass: row-stochastic Q is entrywise nonnegative (**def-stochastic**), so for every entry $\max\{-P_{ij}, 0\} \leq |P_{ij} - Q_{ij}|$ — trivially where $P_{ij} \geq 0$, and because $-P_{ij} \leq Q_{ij} - P_{ij}$ where $P_{ij} < 0$. Summing over j and maximising over i gives, with δ as in **def-negative-mass**,

$$\delta(P) \leq \|P - Q\|_{\infty \rightarrow \infty} \leq 12\eta. \quad (3)$$

So $C = 12$ serves. The export records an instructive scope correction here: the negative-mass step was first blocked because **def-negative-mass** — although permitted by the registry shard — had not been registered inside the workspace, and the tree carried an explicit gap certificate rather than the estimate; registering the definition discharged it.

Converse direction. Let P be an exact signed idempotent with negative mass δ . Split each row into disjointly supported nonnegative parts, $p_i = p_i^+ - p_i^-$, and set $a_i = \sum_k (P_{ik})_- \leq \delta$. Total mass one (**def-signed-idempotent**) gives $\sum_k p_{ik}^+ = 1 + a_i$, so

$$q_i = \frac{p_i^+}{1 + a_i}$$

is a probability vector and the matrix Q with rows q_i is row-stochastic. The correction $e_i = q_i - p_i$ is

$$e_i = p_i^- - \frac{a_i}{1 + a_i} p_i^+,$$

a difference of two disjointly supported vectors of ℓ^1 -mass a_i each; hence $\|e_i\|_1 = 2a_i \leq 2\delta$ and $\|P - Q\|_{\infty \rightarrow \infty} \leq 2\delta$.

For the idempotence defect, write $D = Q - P$. The row-geometry clause of **def-signed-idempotent** gives $\|P\|_{\infty \rightarrow \infty} \leq 1 + 2\delta$, hence $\|P - I\|_{\infty \rightarrow \infty} \leq 2 + 2\delta$. Using $P^2 = P$ exactly, each row satisfies the exact decomposition

$$q_i Q - q_i = e_i(P - I) + q_i D. \quad (4)$$

Since q_i is a probability vector, $\|q_i D\|_1 \leq \|D\|_{\infty \rightarrow \infty} \leq 2\delta$, while $\|e_i(P - I)\|_1 \leq 2\delta(2 + 2\delta)$. Adding, $\|Q^2 - Q\|_{\infty \rightarrow \infty} \leq 6\delta + 4\delta^2$, exactly as contracted.

Status. proved; **af:** validated (T0). Workspace **proofs/lem-classical-equiv**: a 29-node tree (21 live nodes validated, 8 superseded nodes archived), root **validated**, taint clean.

Role in the chain. This is the licence to move between the two pictures of §1, and the only such licence in the registry. It has no dependencies. Everything stated in the signed picture reaches the stochastic formulation of **op-classical** only through it, and only up to the universal constants above; **op-classical** itself remains open.

3 Positive-retract hardening

Lemma 2 (`lem-prh`). *Let $k, n \geq 1$ and let*

$$A : \ell^\infty(k) \longrightarrow \ell^\infty(n), \quad M : \ell^\infty(n) \longrightarrow \ell^\infty(k)$$

be positive unital maps — equivalently, maps whose matrices have probability-vector rows. If

$$\|MA - I_k\|_{\infty \rightarrow \infty} \leq \varepsilon, \quad 0 \leq \varepsilon < \frac{1}{2},$$

*then there is a stochastic idempotent $E : \ell^\infty(n) \rightarrow \ell^\infty(n)$ (**def-stochastic**) with*

$$\|AM - E\|_{\infty \rightarrow \infty} \leq 2\sqrt{2\varepsilon}. \quad (5)$$

Transcription note. The pair (A, M) is the datum of **def-positive-approximate-retract**, with retract defect $\varepsilon_{\text{ret}} = \|MA - I_k\|_{\infty \rightarrow \infty}$. Following the validated tree, A is presented by rows $a_i \in \Delta_k$ indexed by $i \in \{1, \dots, n\}$ and M by rows μ_s indexed by $s \in \{1, \dots, k\}$, so that the s -th row of MA is the probability vector $\sum_i \mu_s(i) a_i$.

Registry contract (`verbatim`).

Positive-retract hardening (PRH): let $k, n \geq 1$ and let $A : \ell^\infty(k) \rightarrow \ell^\infty(n)$ and $M : \ell^\infty(n) \rightarrow \ell^\infty(k)$ be positive unital maps (equivalently, have probability-vector rows); if $\|MA - I_k\|_{\infty \rightarrow \infty} \leq \varepsilon$ with $0 \leq \varepsilon < 1/2$, then there is a stochastic idempotent $E : \ell^\infty(n) \rightarrow \ell^\infty(n)$ with $\|AM - E\|_{\infty \rightarrow \infty} \leq 2\sqrt{2\varepsilon}$.

Proof (prose account of the af-validated tree). The tree splits on the defect. If $\varepsilon = 0$ the hypothesis reads $MA = I_k$ exactly, and then $E := AM$ is already row-stochastic with $E^2 = A(MA)M = AM = E$, so (5) holds with error 0. The substance is the case $0 < \varepsilon < \frac{1}{2}$.

The scale and the cores. Fix

$$\lambda = \sqrt{\varepsilon/2}, \quad C_s = \{i : a_{is} > 1 - \lambda\}, \quad \beta_s = \mu_s(C_s^c).$$

The single quantitative input is a coordinate defect identity. The s -th row of MA is the probability vector $\sum_i \mu_s(i) a_i$, and for probability vectors the ℓ^1 distance to the basis vector e_s is governed by one coordinate:

$$\|(MA)_{s,*} - e_s\|_1 = 2(1 - (MA)_{ss}) = 2 \sum_i \mu_s(i) (1 - a_{is}) =: 2D_s, \quad (6)$$

so the norm hypothesis is exactly $D_s \leq \varepsilon/2$ for every s . Two consequences follow. First, the cores are pairwise disjoint: if $i \in C_s \cap C_t$ with $s \neq t$ then $a_{is} + a_{it} > 2(1 - \lambda) > 1$ (using $\lambda < \frac{1}{2}$, which is where $\varepsilon < \frac{1}{2}$ enters), contradicting that a_i is a probability vector. Second, off the core one has $1 - a_{is} \geq \lambda$, so dropping the on-core terms of the nonnegative sum D_s gives $\lambda\beta_s \leq D_s \leq \varepsilon/2$, whence

$$\beta_s \leq \frac{\varepsilon}{2\lambda} = \lambda < \frac{1}{2}. \quad (7)$$

The tree records a correction at this point: an earlier node asserted the strict inequality $\beta_s < \lambda$, which the threshold $a_{is} \leq 1 - \lambda$ off the core does not give; the weak form (7) is what holds and all that is used.

The exact construction. Condition each decoder row on its core and replace each encoder row inside a core by a basis vector:

$$\nu_s(i) = \frac{\mu_s(i) \mathbf{1}_{C_s}(i)}{1 - \beta_s}, \quad \hat{A} \text{ has row } e_s \text{ on } C_s \text{ and row } a_i \text{ off } \bigcup_s C_s.$$

Both are well defined: $\beta_s < 1$ makes ν_s a probability vector, and disjointness of the cores makes $\hat{A}$ unambiguous. The construction is engineered so that the retraction becomes *exact*: the s -th row of $N\hat{A}$ is $\sum_{i \in C_s} \mu_s(i) e_s / (1 - \beta_s) = e_s$, i.e.

$$N\hat{A} = I_k, \quad \text{hence} \quad E := \hat{A}N \text{ satisfies } E^2 = \hat{A}(N\hat{A})N = \hat{A}N = E. \quad (8)$$

As a product of positive unital maps, E is row-stochastic, so it is a stochastic idempotent in the sense of **def-stochastic**.

The two error terms. Write $AM - E = A(M - N) + (A - \hat{A})N$ and bound the summands separately.

For the conditioning term, ν_s is μ_s conditioned on C_s , so the two rows differ by mass β_s removed from C_s^c and mass β_s redistributed on C_s ; hence $\|\mu_s - \nu_s\|_1 = 2\beta_s \leq \varepsilon/\lambda$ by (7) and $\|M - N\|_{\infty \rightarrow \infty} \leq \varepsilon/\lambda$. Because A has probability rows it is an $\infty \rightarrow \infty$ contraction — concretely, $\sum_j |(AB)_{ij}| \leq \sum_s a_{is} \sum_j |B_{sj}| \leq \|B\|_{\infty \rightarrow \infty}$ — so $\|A(M - N)\|_{\infty \rightarrow \infty} \leq \varepsilon/\lambda$.

For the core-replacement term, the rows of $(A - \hat{A})N$ vanish outside $\bigcup_s C_s$. For $i \in C_s$ the corresponding row is $\sum_t a_{it} \nu_t - \nu_s$, and since every ν_t is a probability vector its ℓ^1 norm is at most $(1 - a_{is}) + \sum_{t \neq s} a_{it} = 2(1 - a_{is}) < 2\lambda$. Hence $\|(A - \hat{A})N\|_{\infty \rightarrow \infty} \leq 2\lambda$.

Optimisation. Adding the two bounds,

$$\|AM - E\|_{\infty \rightarrow \infty} \leq \frac{\varepsilon}{\lambda} + 2\lambda,$$

and the choice $\lambda = \sqrt{\varepsilon/2}$ balances the two terms at the common value $\sqrt{2\varepsilon}$, giving (5). The square-root rate is intrinsic to this balance: conditioning costs ε/λ and core replacement costs 2λ , and no choice of threshold does better than their geometric mean.

Status. proved; af: validated (T0). Workspace **proofs/lem-prh**: 14 nodes, all validated, root validated, taint clean.

Role in the chain. **lem-prh** has no registry dependencies. It is the finishing step of the live route: it installs the reduction of **op-classical** to the *existence* of a positive approximate retract for a given almost-idempotent Q — stochastic A, M with $\|AM - Q\|_{\infty \rightarrow \infty} = O(\eta)$ and $\|MA - I\|_{\infty \rightarrow \infty} = O(\eta)$ — after which (5) delivers the exact idempotent at rate $\sqrt{\eta}$. Producing such a pair is what the COMP and H-CB tiers below are ultimately for, and that production is *not* established: **lem-prh** is a conditional finisher, and **op-classical** remains open. Downstream registry consumers are **lem-route-f-prh-finish** and **lem-route-f-k-ledger**, both non-rigorous.

4 Amplification naturality of the functional calculus

Lemma 3 (`lem-compcb-amplification-naturality`). *Let $\mathcal{B}$ be a unital Banach algebra and let $\iota_n : \mathcal{B} \rightarrow M_n(\mathcal{B})$, $X \mapsto 1_{M_n} \otimes X$, be the unital amplification, an isometric unital homomorphism. Let f be given on $\|X - x_0 I\| < r$ by the power series*

$$f(X) = a_0 I + \sum_{m \geq 1} a_m (X - x_0 I)^m. \quad (9)$$

Then for every $X \in \mathcal{B}$ with $\|X - x_0 I\| < r$ the element $f(\iota_n(X))$ is defined and

$$f(\iota_n(X)) = \iota_n(f(X)). \quad (10)$$

In particular, whenever $\|(2P - I)^2 - I\| < 1$,

$$\theta(\iota_n(2P - I)) = \iota_n(\theta(2P - I)), \quad \theta(X) = \frac{1}{2}(I + \operatorname{sgn} X).$$

Transcription note. θ and sgn are the power-series functional calculus of `def-theta-idempotent-approximation`; only its definitional displays are in scope here. The norm is the ambient Banach-algebra norm.

Registry contract (`verbatim`).

Amplification naturality of the power-series functional calculus: let B be a unital Banach algebra, ι_n : $B \rightarrow M_n(B)$ the unital amplification $X \mapsto 1_{\{M_n\}}$ tensor X (an isometric unital homomorphism), and f given on $\|X - x_0 I\| < r$ by the power series $f(X) = a_0 I + \sum_{m \geq 1} a_m (X - x_0 I)^m$; then for every X in B with $\|X - x_0 I\| < r$, $f(\iota_n(X))$ is defined and $f(\iota_n(X)) = \iota_n(f(X))$; in particular, whenever $\|(2P - I)^2 - I\| < 1$, $\theta(\iota_n(2P - I)) = \iota_n(\theta(2P - I))$ with $\theta(X) = (I + \operatorname{sgn}(X))/2$.

Proof (prose account of the af-validated tree). *The general statement.* Three structural facts about ι_n do all the work. Unitality gives $\iota_n(I_B) = I_{M_n(\mathcal{B})}$, hence $\iota_n(X) - x_0 I = \iota_n(X - x_0 I)$; isometry then transports the convergence hypothesis verbatim, $\|\iota_n(X) - x_0 I\| = \|X - x_0 I\| < r$, so the series (9) evaluated at $\iota_n(X)$ is inside its domain; and multiplicativity gives $(\iota_n(X) - x_0 I)^m = \iota_n((X - x_0 I)^m)$ for every $m \geq 0$. Consequently the N -th partial sums correspond exactly,

$$T_N = a_0 I + \sum_{m=1}^N a_m (\iota_n(X) - x_0 I)^m = \iota_n(S_N), \quad S_N = a_0 I + \sum_{m=1}^N a_m (X - x_0 I)^m.$$

Since $S_N \rightarrow f(X)$ in $\mathcal{B}$ and ι_n is isometric, hence continuous, $T_N \rightarrow \iota_n(f(X))$. The power-series definition identifies the limit of the same sequence T_N with $f(\iota_n(X))$, so uniqueness of limits yields both definedness and (10). No property of ι_n beyond continuity, unitality and multiplicativity is used.

The θ specialization. Here the functional calculus is not a single series but a composite built from the inverse square root, so the tree first pins that branch down. With $c_m = \binom{-1/2}{m}$ one has $c_0 = 1$ and $c_m/c_{m-1} = -(2m-1)/(2m)$, whose modulus tends to 1; the ratio test therefore gives radius exactly one for $h(w) = \sum_{m \geq 0} c_m w^m$. On $|w| < 1$ termwise differentiation is legitimate and the coefficient recurrence is precisely the differential equation $2(1+w)h'(w) + h(w) = 0$; hence $\frac{d}{dw} [(1+w)h(w)^2] = 0$, and $h(0) = 1$ gives $(1+w)h(w)^2 = 1$. So h is the analytic inverse-square-root branch normalised at 1.

Transporting this to a unital Banach algebra: for $\|Z - I\| < 1$ the series $H = \sum_{m \geq 0} c_m (Z - I)^m$ converges absolutely in norm, and absolute convergence licenses the same Cauchy products as in the scalar identity, so $ZH^2 = I$; the normalisation $H = I + (\text{terms of positive degree})$ identifies H with the branch $Z^{-1/2}$ used by the registered calculus. Applying the general statement to this series with $Z = Y^2$, $Y = 2P - I$, gives $(\iota_n(Y)^2)^{-1/2} = \iota_n((Y^2)^{-1/2})$ whenever $\|Y^2 - I\| < 1$; multiplicativity then propagates the identity through $\text{sgn}(Y) = Y(Y^2)^{-1/2}$ and finally through $\theta = \frac{1}{2}(I + \text{sgn})$.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-amplification-naturality: 12 nodes, all validated, root validated, taint clean.

Role in the chain. This lemma exists because of a stall, and the shape of the campaign record is worth stating: three separate blocking challenges in the first orchestration of `lem-compcb-amplified-compression` named this one identity, so it was factored out into its own shard and validated first, bottom-up. It has no dependencies. It is imported by `lem-compcb-amplified-compression` (§5) and by `lem-compcb-entrywise-compression-naturality` (§7), and it is the reason every compression identity in the COMP tier holds at *every* amplification level rather than only at level one.

5 The amplified compression identity

Lemma 4 (`lem-compcb-amplified-compression`). *There is a universal $e_{\text{cmp}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{cmp}}$, every pair of δ -projections P, Q in an extended ε - C^* -algebra and every $n \geq 1$ satisfy*

$$1_{M_n} \otimes \text{Co}_{P,Q} = \text{Co}_{I_n \otimes P, I_n \otimes Q} \quad (11)$$

and

$$M_n \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{I_n \otimes P, I_n \otimes Q}. \quad (12)$$

Transcription note. $\text{Co}_{P,Q}$ and $\mathcal{S}_{P,Q}$ are the compression map and compressed corner of **def-compressed-corner**; δ -projections are as in **def-delta-projection**, and “extended” is as in **def-extended-epsilon-cstar-algebra**, so that every $M_n \otimes \mathcal{A}$ is itself an ε - C^* -algebra. On the left of (11) $1_{M_n} \otimes (\cdot)$ denotes the entrywise amplification of an operator on $\mathcal{A}$; on the right $I_n \otimes P$ is the amplified element.

Registry contract (`verbatim`).

Amplified compression identity: there is a universal $e_{\text{cmp}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{cmp}}$, every pair of δ -projections P, Q in an extended ε - C^* -algebra and every $n \geq 1$ satisfy $1_{\{M_n\}} \text{ tensor } \text{Co}_{\{P,Q\}} = \text{Co}_{\{I_n \text{ tensor } P, I_n \text{ tensor } Q\}}$ and $M_n \text{ tensor } \mathcal{S}_{\{P,Q\}} = \mathcal{S}_{\{I_n \text{ tensor } P, I_n \text{ tensor } Q\}}$.

Proof (prose account of the af-validated tree). *A uniform threshold.* Write $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Because $\mathcal{A}$ is *extended*, each $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra with compatible matrix multiplication, involution, unit and amplification norm; since $P_n^\dagger = P_n$ and $\|P_n^2 - P_n\| = \|I_n \otimes (P^2 - P)\| = \|P^2 - P\| \leq \delta$, the amplified elements are δ -projections *with the same δ , uniformly in n* . Next, the compression is the value of θ at a fixed operator: writing $A_{P,Q} = \frac{1}{2}(L_P R_Q + R_Q L_P)$ and $T_{P,Q} = 2A_{P,Q} - I$, **def-compressed-corner** supplies a single universal C with $\|A_{P,Q}^2 - A_{P,Q}\| \leq Ce$ in *every* ε - C^* -algebra, and the exact operator identity

$$T_{P,Q}^2 - I = 4(A_{P,Q}^2 - A_{P,Q}) \quad (13)$$

converts this into $\|T_{P,Q}^2 - I\| \leq 4Ce$. Since the same C applies inside every $M_n \otimes \mathcal{A}$ to the pair (P_n, Q_n) , choosing e_{cmp} with $4Ce_{\text{cmp}} < 1$ puts $T_{P,Q}$ and every T_{P_n, Q_n} simultaneously inside the θ/sgn domain of **def-theta-idempotent-approximation**. Uniformity of the threshold in n is exactly what (13) buys.

The amplification is entrywise, not diagonal. Left and right multiplication by the block-diagonal P_n, Q_n act slotwise: for $X = [x_{ij}]$ one has $L_{P_n} R_{Q_n}(X) = [P(x_{ij}Q)]$ and $R_{Q_n} L_{P_n}(X) = [(Px_{ij})Q]$, whence $T_{P_n, Q_n} = 1_{M_n} \otimes T_{P,Q}$. Here the tree records a correction that is genuinely load-bearing. The relevant map is *not* the diagonal embedding of $B(\mathcal{A})$ into $M_n(B(\mathcal{A}))$ but the entrywise operator amplification

$$\Gamma_n : B(\mathcal{A}) \rightarrow B(M_n \otimes \mathcal{A}), \quad \Gamma_n(U)([x_{ij}]) = [U(x_{ij})],$$

and the tree proves exactly the properties Γ_n has and no more: coordinate compression gives $\|x_{ij}\| \leq \|X\|$ and $\|E_{ij} \otimes y\| = \|y\|$, so $\|\Gamma_n(U)\| \leq n^2 \|U\|$ — bounded at each fixed n , with *no* claim of isometry and *no* uniform-in- n operator bound; and entrywise evaluation gives $\Gamma_n(I) = I$ and $\Gamma_n(UV) = \Gamma_n(U)\Gamma_n(V)$. Boundedness at fixed n is all that is needed, because the θ series is a norm limit for each fixed n separately.

Transporting θ . Put $T = T_{P,Q}$ and $R = T^2 - I$. On $\|R\| < 1$ the registered calculus writes $\theta(T) = \frac{1}{2}(I + T \sum_{k \geq 0} b_k R^k)$ with b_k the Taylor coefficients of $(1 + z)^{-1/2}$ — the branch identified in `lem-compcb-amplification-naturality`. Since Γ_n is a unital algebra homomorphism it commutes with every partial sum and satisfies $\Gamma_n(R) = \Gamma_n(T)^2 - I$; boundedness lets one pass to the norm limit; and the threshold above guarantees that the limit at $\Gamma_n(T) = T_{P_n, Q_n}$ is the value $\theta(T_{P_n, Q_n})$. Hence $\Gamma_n(\theta(T_{P,Q})) = \theta(T_{P_n, Q_n})$, which by $\text{Co}_{P,Q} = \theta(T_{P,Q})$ is precisely (11).

Ranges. For any linear $C : \mathcal{A} \rightarrow \mathcal{A}$ and finite n one has $\text{Im}(1_{M_n} \otimes C) = M_n \otimes \text{Im } C$: every output has entries $C(x_{ij})$, and conversely finitely many entrywise preimages assemble into a single matrix preimage. Since $\mathcal{S}_{P,Q} = \text{Im } \text{Co}_{P,Q}$ by `def-compressed-corner`, applying this to (11) gives (12).

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-amplified-compression: 13 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-amplification-naturality`. This is the structural key-stone of the COMP tier: every later result that says “uniformly in n ” does so because (11)–(12) let a level-one fact be read at level n with no loss. It is imported by `lem-compcb-amplified-compression-identities`, `lem-compcb-entrywise-compression-naturality`, `lem-compcb-rectangular-product`, `lem-compcb-amplified-compression-identities`, `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-compcb-single-compression-transfer` and `lem-hcb2-product-defect`.

6 Idempotence and adjoint symmetry of amplified compressions

Lemma 5 (`lem-compcb-amplified-compression-identities`). *There is a universal $e_{\text{cmp}} > 0$ such that, whenever $\mathcal{A}$ is an extended ε - C^* -algebra, $e = \delta + \varepsilon \leq e_{\text{cmp}}$, P, Q are δ -projections in $\mathcal{A}$, $n \geq 1$, and $X \in M_n \otimes \mathcal{A}$, one has*

$$\text{Co}_{P_n, Q_n}^2 = \text{Co}_{P_n, Q_n}, \quad \text{Co}_{P_n, Q_n}(X)^\dagger = \text{Co}_{Q_n, P_n}(X^\dagger), \quad (14)$$

where $P_n = I_n \otimes P$ and $Q_n = I_n \otimes Q$.

Registry contract (verbatim).

Amplified compression identities: there is a universal $e_{\text{cmp}} > 0$ such that, whenever $\mathcal{A}$ is an extended ε - C^* -algebra, $e = \delta + \varepsilon \leq e_{\text{cmp}}$, P, Q are δ -projections in $\mathcal{A}$, $n \geq 1$, and X is in M_n tensor $\mathcal{A}$, one has $\text{Co}_{\{P_n, Q_n\}}^2 = \text{Co}_{\{P_n, Q_n\}}$ and $\text{Co}_{\{P_n, Q_n\}}(X)^\dagger = \text{Co}_{\{Q_n, P_n\}}(X^\dagger)$, where $P_n = I_n$ tensor P and $Q_n = I_n$ tensor Q .

Proof (prose account of the af-validated tree). The argument is short because the amplification lemma has already done the structural work; what remains is to check that two *exact* level-one identities survive the passage to matrices.

Identification. Take e_{cmp} to be the universal constant supplied by `lem-compcb-amplified-compression`. For $e \leq e_{\text{cmp}}$ that lemma gives $\text{Co}_{P_n, Q_n} = 1_{M_n} \otimes \text{Co}_{P, Q}$, and — applied a second time to the *ordered* pair (Q, P) , which is a distinct instance — also $\text{Co}_{Q_n, P_n} = 1_{M_n} \otimes \text{Co}_{Q, P}$. Both amplified compressions are therefore canonical amplifications of level-one maps.

The level-one identities. Writing $T = \text{Co}_{P, Q}$ and $U = \text{Co}_{Q, P}$, **def-compressed-corner** states exactly $T^2 = T$ and $T(x)^\dagger = U(x^\dagger)$ for all $x \in \mathcal{A}$. These are equalities, not estimates; no smallness is consumed by them, and the threshold above is needed only for the identification step.

Entrywise transfer. Let T, U be any linear maps on a self-adjoint operator space satisfying those two identities, and let $T_n = 1_{M_n} \otimes T$, $U_n = 1_{M_n} \otimes U$. For $X = [x_{ij}]$, idempotence is checked slotwise,

$$(T_n^2 X)_{ij} = T(T(x_{ij})) = (T^2)(x_{ij}) = T(x_{ij}) = (T_n X)_{ij},$$

so $T_n^2 = T_n$. Adjoint symmetry combines the matrix involution with the slotwise action: the (i, j) entry of $T_n(X)^\dagger$ is $T(x_{ji})^\dagger = U(x_{ji}^\dagger)$, while the (i, j) entry of $U_n(X^\dagger)$ is $U((X^\dagger)_{ij}) = U(x_{ji}^\dagger)$. The transposition of indices is what makes the *swap* $\text{Co}_{P_n, Q_n} \mapsto \text{Co}_{Q_n, P_n}$ in (14) the right statement. Combining the identification with the transfer gives the lemma.

A recorded contract amendment. The export’s final node is a contract-reconciliation step, and it documents why the statement is shaped as it is. The originally codified contract left the ambient algebra unbound: a challenge observed that the proof used, but the contract never asserted, that $\mathcal{A}$ is an extended ε - C^* -algebra with P, Q in $\mathcal{A}$. The contract was amended to the prover’s corrected root text — the version reproduced above — as a mechanical, verdict-driven typing repair. The added clauses are the intended and already-used hypotheses of the source argument, so nothing was strengthened or weakened.

Status. proved; af: validated (T0). Workspace `proofs/lem-compcb-amplified-compression-identities`: 7 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression`. These two identities are the algebraic skeleton on which every quantitative COMP estimate is hung: idempotence is what makes “ X lies in the corner” equivalent to “ X is fixed by its compression”, and the dagger swap

is what lets an estimate proved on one side of a rectangular corner be read on the other. It is imported by `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-unit-norm`, `lem-compcb-compressed-unit-action`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`, `lem-hcb2-amplified-adjointness` and `lem-hcb1-variational-`

7 Entrywise compression naturality at a matrix unit

Lemma 6 (`lem-compcb-entrywise-compression-naturality`). *There is a universal $e_{\text{nat}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{nat}}$, Q is a δ -projection in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and E_{11} is the $(1,1)$ matrix unit of M_n , every $Z \in \mathcal{A}$ satisfies*

$$\text{Co}_{I_n \otimes Q, I_n \otimes Q}(E_{11} \otimes Z) = E_{11} \otimes \text{Co}_Q(Z). \quad (15)$$

Transcription note. Co_Q is the $P = Q$ abbreviation $\text{Co}_{Q,Q}$ of **def-compressed-corner**. The point of the statement is that the amplified compression does not merely preserve the corner but acts in the single occupied slot, leaving the zero blocks zero.

Registry contract (verbatim).

Entrywise compression naturality: there is a universal $e_{\text{nat}} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{\text{nat}}$, Q is a δ -projection in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and E_{11} is the $(1,1)$ matrix unit of M_n , every Z in $\mathcal{A}$ satisfies $\text{Co}_{I_n \otimes Q, I_n \otimes Q}(E_{11} \otimes Z) = E_{11} \otimes \text{Co}_Q(Z)$.

Proof (prose account of the af-validated tree). Set $e_{\text{nat}} := e_{\text{cmp}}$, the universal threshold supplied by `lem-compcb-amplified-compression`. That lemma is an equality of *linear maps* on $M_n(\mathcal{A})$, and the proof consists of specializing it and then evaluating.

Specializing the imported identity to the pair $(P, Q) = (Q, Q)$ gives

$$\text{Co}_{I_n \otimes Q, I_n \otimes Q} = 1_{M_n} \otimes \text{Co}_{Q,Q} = 1_{M_n} \otimes \text{Co}_Q,$$

the last equality being nothing but the abbreviation $\text{Co}_{Q,Q} = \text{Co}_Q$ recorded in **def-compressed-corner**. This is where the whole quantitative content of the statement sits: the threshold, the θ -domain control and the amplification argument are all inherited from `lem-compcb-amplified-compression`, which in turn rests on `lem-compcb-amplification-naturality`.

What remains is a one-line evaluation. For any linear $T : \mathcal{A} \rightarrow \mathcal{A}$ the tensor amplification acts on elementary tensors by $(1_{M_n} \otimes T)(X \otimes Z) = X \otimes T(Z)$; taking $T = \text{Co}_Q$ and $X = E_{11}$ gives (15).

Status. `proved; af: validated (T0)`. Workspace `proofs/lem-compcb-entrywise-compression-naturality`: 4 nodes, all `validated`, root `validated`, taint `clean`.

Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplification-naturality`. Like §4, this shard exists because of a recorded stall: the column-Hilbert workspace (§16) reached its third consecutive stall on exactly the identity (15), whose statement was then extracted mechanically from the blocking challenge and validated on its own. It is a strictly sharper statement than the $1_{M_n} \otimes$ -clause it specializes — that clause identifies two maps, whereas (15) asserts what happens to a *single-slot* element — and that sharper form is what the column-Hilbert estimate consumes. It has one registry consumer, `lem-hcb-column-hilbert-squared`.

8 The uniform rectangular compressed-product estimate

Lemma 7 (`lem-compcb-rectangular-product`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular pair A, B satisfies*

$$\|A \cdot B - AB\| \leq C_{\text{co}} e \|A\| \|B\|. \quad (16)$$

Transcription note. The tree fixes “compatible amplified rectangular pair” as an explicit abbreviation, and the quantifier in (16) ranges over exactly these data and no others: an extended ε - C^ -algebra $\mathcal{A}$; an integer $n \geq 1$; δ -projections P, Q, R in $\mathcal{A}$; the amplifications $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$, $R_n = I_n \otimes R$; elements $A \in \mathcal{S}_{P_n, Q_n}$ and $B \in \mathcal{S}_{Q_n, R_n}$; and the compressed product $A \cdot B = \text{Co}_{P_n, R_n}(AB)$ formed as in **def-compressed-corner** inside $M_n \otimes \mathcal{A}$. In particular both factors live in one square amplification $M_n \otimes \mathcal{A}$ — a restriction that later forces a separate row/column statement (§13).*

Registry contract (`verbatim`).

Uniform rectangular compressed-product estimate: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular pair satisfies $\|A \cdot B - AB\| \leq C_{\text{co}} * e * \|A\| * \|B\|$.

Proof (prose account of the af-validated tree). The shape of the argument is: unpack a level-one universal estimate into named constants, verify that the amplified data satisfy its hypotheses, and apply it inside $M_n \otimes \mathcal{A}$. No new analysis is done at level n ; the whole point is that none is needed.

Unpacking the level-one estimate. **def-compressed-corner** states, for any triple U, V, W of δ -projections in an ε - C^* -algebra, that $X \cdot Y = \text{Co}_{U, W}(XY)$ on $\mathcal{S}_{U, V} \times \mathcal{S}_{V, W}$ and that $\|X \cdot Y - XY\| \leq O(\delta + \varepsilon) \|X\| \|Y\|$. The tree makes the big- O explicit: there are constants $C_{\text{corner}} < \infty$ and $e_{\text{corner}} > 0$, independent of the algebra, of the projections and of the elements, such that the bound reads $C_{\text{corner}} e \|X\| \|Y\|$ whenever $e \leq e_{\text{corner}}$. This independence is the reason the final constant is uniform in n : the level-one estimate is a statement about *every* ε - C^* -algebra, and $M_n \otimes \mathcal{A}$ is one of them.

The amplified data are admissible. Because $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra for every n ; and P_n is a δ -projection whenever P is, since $P_n^\dagger = P_n$, $P_n^2 - P_n = I_n \otimes (P^2 - P)$ and the block-diagonal norm satisfies $\|I_n \otimes Z\| = \|Z\|$ — likewise for Q_n, R_n . Applying **lem-compcb-amplified-compression** to the three pairs (P, Q) , (Q, R) , (P, R) identifies the amplified corners with $M_n \otimes \mathcal{S}_{P, Q}$, $M_n \otimes \mathcal{S}_{Q, R}$, $M_n \otimes \mathcal{S}_{P, R}$ and the amplified compressions with the entrywise amplifications; **lem-compcb-amplified-compression** supplies the idempotence and dagger identities for those maps. Hence a compatible amplified rectangular pair really is an ordinary rectangular pair in $M_n \otimes \mathcal{A}$, and its amplified compressed product is literally the ordinary compressed product there, $A \cdot B = \text{Co}_{P_n, R_n}(AB)$, which lands in $\mathcal{S}_{P_n, R_n}$.

Assembly. Take

$$C_{\text{co}} = C_{\text{corner}}, \quad e_{\text{co}} = \min(e_{\text{corner}}, e_{\text{cmp}}),$$

with e_{cmp} the threshold of the amplification dependencies; both are universal, finite and positive. For $e \leq e_{\text{co}}$ the level-one estimate applies inside $M_n \otimes \mathcal{A}$ with $U = P_n$, $V = Q_n$, $W = R_n$, $X = A$, $Y = B$, and gives (16) with constants independent of n and of all the data.

Status. `proved`; `af`: `validated (T0)`. Workspace `proofs/lem-compcb-rectangular-product`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression` and `lem-compcb-amplified-compression-i`. This is the workhorse estimate of the COMP tier — the licence to replace a compressed product by an ambient product at cost $O(e)$ — and it is invoked by almost everything downstream: `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-hcb0-compressed-associator`, `lem-compcb-corner-algebra`, `lem-hcb-column-hilbert-squared`, `lem-compcb-single-compression-transfer`, and `lem-hcb2-product-defect`. Its square-amplification compatibility clause is a real restriction, not a formality: §13 exists precisely because a row/column pair fails it.

9 Amplified almost-containment of corners

Lemma 8 (`lem-compcb-amplified-almost-containment`). *There are universal $C_{ac} < \infty$ and $e_{ac} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ac}$, the elements P_1, P, Q_1, Q are δ -projections with*

$$\|P_1 P - P_1\| \leq \delta, \quad \|Q_1 Q - Q_1\| \leq \delta, \quad (17)$$

and $n \geq 1$, every $X \in M_n \otimes \mathcal{S}_{P_1, Q_1}$ satisfies

$$\|\text{Co}_{P_n, Q_n}(X) - X\| \leq C_{ac} e \|X\|, \quad (18)$$

with $P_n = I_n \otimes P$, $Q_n = I_n \otimes Q$. Transcription note. Hypothesis (17) is the approximate form of “ $P_1 \leq P$ and $Q_1 \leq Q$ ”; the conclusion says that the inner corner $\mathcal{S}_{P_1, Q_1}$ sits almost inside the outer one, uniformly at every amplification level.

Registry contract (`verbatim`).

Amplified almost-containment: there are universal $C_{ac} < \text{infinity}$ and $e_{ac} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ac}$, P_1, P, Q_1, Q are δ -projections with $\|P_1 P - P_1\|, \|Q_1 Q - Q_1\| \leq \delta$, every $n \geq 1$, and X in $M_n \otimes \mathcal{S}_{P_1, Q_1}$, one has $\|\text{Co}_{P_n, Q_n}(X) - X\| \leq C_{ac} e \|X\|$.

Proof (prose account of the af-validated tree). *A uniform ledger.* All four elements amplify to δ -projections in $M_n \otimes \mathcal{A}$, and the containment hypotheses amplify too, with the involution converting one side into the other: $P_n P_{1,n} - P_{1,n} = I_n \otimes (P P_1 - P_1) = I_n \otimes (P_1 P - P_1)^\dagger$ has norm at most δ , and $Q_{1,n} Q_n - Q_{1,n} = I_n \otimes (Q_1 Q - Q_1)$ likewise. A crude but uniform norm bound is also available: for $\varepsilon, \delta \leq \frac{1}{2}$ the ε - C^* axioms and `def-delta-projection` give $(1 - \varepsilon)\|R_n\|^2 \leq \|R_n^2\| \leq \|R_n\| + \delta$, hence $\|R_n\| \leq B := 3$ for each of the four. Finally `def-compressed-corner` supplies one universal k_{co} with

$$\|\text{Co}_{R,S}(Z) - R(ZS)\| \leq k_{co} e \|Z\|, \quad \|\text{Co}_{R,S}(Z) - (RZ)S\| \leq k_{co} e \|Z\|, \quad (19)$$

in every $M_n \otimes \mathcal{A}$. The threshold e_0 is the minimum of $\frac{1}{2}$, the threshold behind (19), and those of the two amplification dependencies.

Control in the source corner. By `lem-compcb-amplified-compression`, $M_n \otimes \mathcal{S}_{P_1, Q_1} = \mathcal{S}_{P_{1,n}, Q_{1,n}}$, so X lies in the image of $\text{Co}_{P_{1,n}, Q_{1,n}}$; idempotence (`lem-compcb-amplified-compression-identities`) then upgrades membership to the fixed-point identity $\text{Co}_{P_{1,n}, Q_{1,n}}(X) = X$. Feeding this into (19) with $R = P_{1,n}$, $S = Q_{1,n}$, $Z = X$ replaces the compression by X itself and yields the two bracketings

$$\|X - P_{1,n}(XQ_{1,n})\| \leq k_{co} e \|X\|, \quad \|X - (P_{1,n}X)Q_{1,n}\| \leq k_{co} e \|X\|. \quad (20)$$

Transfer to the outer projections. The claim is that X is almost invariant under P_n on the left and Q_n on the right, with a universal $L := k_{co}(2B + 1) + 2B(B^2 + 2)$. For the left estimate put $D = P_{1,n}(XQ_{1,n})$ and $Y = XQ_{1,n}$, and telescope

$$P_n X - X = P_n(X - D) + (P_n D - D) + (D - X).$$

The outer terms are controlled by (20) and submultiplicativity. The middle term is where the containment hypothesis and the *approximate* associativity of an ε - C^* -algebra both enter: $P_n D - D = P_n(P_{1,n}Y) - P_{1,n}Y$ is bounded by $[\varepsilon B^2 + (1 + \varepsilon)\delta]\|Y\|$, since re-associating costs $\varepsilon\|P_n\|\|P_{1,n}\|\|Y\|$

and $(P_n P_{1,n} - P_{1,n})Y$ costs $(1 + \varepsilon)\delta\|Y\|$. With $\|Y\| \leq 2B\|X\|$ and $\varepsilon, \delta \leq e$ the three terms sum to at most $Le\|X\|$. The right estimate is the mirror image, run with $D = (P_{1,n}X)Q_{1,n}$, $Y = P_{1,n}X$ and the hypothesis $\|Q_{1,n}Q_n - Q_{1,n}\| \leq \delta$.

Assembly. Insert $P_n(XQ_n)$ and P_nX :

$$\|\text{Co}_{P_n, Q_n}(X) - X\| \leq \|\text{Co}_{P_n, Q_n}(X) - P_n(XQ_n)\| + \|P_n(XQ_n - X)\| + \|P_nX - X\|.$$

The first term is $\leq k_{co}e\|X\|$ by (19); the second is $\leq (1 + \varepsilon)\|P_n\|Le\|X\| \leq 2BLE\|X\|$; the third is $\leq Le\|X\|$. Hence (18) holds with $C_{ac} = k_{co} + (2B + 1)L$ and $e_{ac} = e_0$, both finite, positive and universal.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-amplified-almost-containment: 13 nodes, all validated, root validated, taint clean.

Role in the chain. Imports lem-compcb-amplified-compression and lem-compcb-amplified-compression-i

It is the corner-inclusion primitive of the COMP tier: it says that once a corner is approximately contained in another, the outer compression is a near-identity on it, again with one constant at every level. Its registry consumer is lem-compcb-single-compression-transfer (§17).

10 The compressed-unit norm estimate

Lemma 9 (`lem-compcb-compressed-unit-norm`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$,*

$$\|u_T\| \leq 1 + C_{\text{co}} e \quad \text{for every } \delta\text{-projection } T, \quad (21)$$

and

$$|\|u_T\| - 1| \leq C_{\text{co}} e \quad \text{for every nonvanishing } T. \quad (22)$$

Transcription note. $u_T = \text{Co}_T(T)$ is the compressed unit of **def-compressed-corner**; “nonvanishing” is the second branch of the alternative in **def-delta-projection**. The asymmetry is deliberate and load-bearing: the two-sided bound is false in general, since a δ -projection may fall in the small branch $\|T\| = O(\delta)$, in which case u_T is small rather than close to a unit.

Registry contract (verbatim).

Compressed-unit norm estimate: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every δ -projection T satisfies $\|u_T\| \leq 1 + C_{\text{co}} * e$, and every nonvanishing T satisfies $\text{abs}(\|u_T\| - 1) \leq C_{\text{co}} * e$.

Proof (prose account of the af-validated tree). This shard’s tree is instructive twice over, because its first attempt was audited and found unsound, and the audit — not the estimate — is what is validated in the early nodes.

The blocked first attempt. The originally registered inputs (**def-extended-epsilon-cstar-algebra** together with **def-compressed-corner**) yield only the δ -projection dichotomy, the identity $u_T = \text{Co}_T(T)$, and the comparison $\|\text{Co}_T - L_T R_T\| \leq ke$. They do *not* yield the quantitative multiplication bound $\|XY\| \leq (1 + \varepsilon)\|X\|\|Y\|$ — the extended definition says only that each $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra, without supplying that term’s quantitative axioms — nor any unit-norm axiom that would give (22) from “ $\mathcal{S}_T$ is an $O(e)$ - C^* -algebra with unit u_T ”. A closure audit node records this verdict explicitly and refuses the root: without those axioms there is no sound bridge from the permitted premises to the contract.

The repair. The missing quantitative content was then registered in the workspace as byte-matched externals — the ε - C^* norm axioms (**epsilon-banach-cstar-norm-axioms**) and the nonvanishing alternative — and each $O(\cdot)$ in them unpacked into a named universal constant after shrinking the threshold. That gives universal $a, b, k < \infty$ and $e_0 > 0$ such that for $e \leq e_0$: (i) every δ -projection T satisfies $\|T\| \leq a\delta$ or is nonvanishing; (ii) $\|XY\| \leq (1 + \varepsilon)\|X\|\|Y\|$; (iii) $u_T = \text{Co}_T(T)$ with $\|\text{Co}_T - L_T R_T\| \leq ke$; and (iv) nonvanishing T obeys $|\|T\| - 1| \leq be$.

Small branch. Suppose $\|T\| \leq a\delta$ and $e \leq \min(e_0, 1)$. Since $(L_T R_T)(T) = T(T^2)$, two applications of (ii) give $\|(L_T R_T)(T)\| \leq (1 + \varepsilon)^2 \|T\|^3 \leq 4a^3 e^3 \leq 4a^3 e$, while (iii) gives $\|u_T - (L_T R_T)(T)\| \leq ke\|T\| \leq kae^2 \leq kae$. Hence $\|u_T\| \leq De$ with $D = 4a^3 + ka$, and in particular $\|u_T\| \leq 1 + De$. Note that here the compressed unit is *small*, which is why no two-sided statement can hold.

Nonvanishing branch. Shrink e_0 so that $e_0 \leq 1$ and $be_0 \leq 1$; then (iv) gives $\|T\| \leq 2$. The compressed unit is compared directly with T rather than through any unit axiom: bilinearity gives $T(T^2) - T = T(T^2 - T) + (T^2 - T)$, so by (iii), (ii) and $\|T^2 - T\| \leq \delta$,

$$\|u_T - T\| \leq \|u_T - (L_T R_T)(T)\| + \|T(T^2) - T\| \leq ke\|T\| + (1 + \varepsilon)\|T\|\delta + \delta \leq (2k + 5)e.$$

The reverse triangle inequality together with (iv) then gives $|\|u_T\| - 1| \leq (2k + 5 + b)e$, hence also the upper bound.

Assembly. With e_{co} the shrunk threshold and $C_{\text{co}} = \max(4a^3 + ka, 2k + 5 + b)$, the dichotomy routes every δ -projection to one of the two branches: the first gives (21), the second gives (21) and (22). The final assembly node states expressly that it uses neither the superseded audit node nor any compressed-unit norm axiom.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-compressed-unit-norm: 15 nodes (12 validated, 3 superseded nodes archived), root validated, taint clean.

Role in the chain. Imports lem-compcb-amplified-compression and lem-compcb-amplified-compression-i. It supplies the unit-size input to lem-compcb-corner-algebra (§15) and to lem-hcb-column-hilbert-squared (§16); in the latter it is applied to the one-dimensional projection Q of an H-CB datum, which is why the nonvanishing hypothesis of (22) has to be discharged there by a separate bridge rather than assumed.

11 Exact adjointness of the amplified Ha map

Lemma 10 (`lem-hcb2-amplified-adjointness`). *There is a universal $e_{\text{adj}} > 0$ such that every H-CB datum with $e \leq e_{\text{adj}}$, every $n \geq 1$ and every $Z \in M_n \otimes \mathcal{S}_{P,R}$ satisfy*

$$(\text{Ha}_{P,R}^Q)_n(Z)^\dagger = (\text{Ha}_{R,P}^Q)_n(Z^\dagger). \quad (23)$$

Transcription note. The datum is that of `def-hcb-datum` and $\text{Ha}_{P,R}^Q$ is the map of `def-ha-map`; $e = \delta + \varepsilon$. The two daggers in (23) are different operations: on the left it is the Hilbert-space adjoint of the operator $(\text{Ha}_{P,R}^Q)_n(Z)$ between amplified column spaces, on the right it is the involution of the ambient algebra applied to Z . The identity is exact — no $O(e)$ error — and the threshold is needed only so that the sesquilinear forms involved are genuine inner products.

Registry contract (`verbatim`).

Exact amplified Ha adjointness: there is a universal $e_{\text{adj}} > 0$ such that every H-CB datum with $e \leq e_{\text{adj}}$, every $n \geq 1$, and every Z in M_n tensor $\mathcal{S}_{\{P,R\}}$ satisfy $(\text{Ha}^Q_{P,R})_n(Z)^\dagger = (\text{Ha}^Q_{R,P})_n(Z^\dagger)$.

Proof (prose account of the af-validated tree). *Threshold.* Set $e_{\text{adj}} = \min(e_{\text{cmp}}, e_{\text{ip}})$, where e_{cmp} comes from `lem-compcb-amplified-compression-identities` and e_{ip} is a universal smallness threshold below which the Euclidean sesquilinear forms of `def-ha-map` and of the column-Hilbert displays are genuine inner products — such a threshold exists because those forms agree with the norms up to a $1 \pm O(e)$ factor. Fix an arbitrary datum, n and Z .

Corner symmetries. Applying `lem-compcb-amplified-compression-identities` at $n = 1$: if $a \in \mathcal{S}_{A,B} = \text{Im Co}_{A,B}$ then $a = \text{Co}_{A,B}(a)$, so $a^\dagger = \text{Co}_{B,A}(a^\dagger)$ and $a^\dagger \in \mathcal{S}_{B,A}$; and since $Q^\dagger = Q$, the compressed unit $\tilde{Q} = \text{Co}_Q(Q)$ is Hermitian.

Exact product reversal. The pivotal algebraic fact is

$$(a \cdot b)^\dagger = b^\dagger \cdot a^\dagger \quad (a \in \mathcal{S}_{A,B}, b \in \mathcal{S}_{B,C}), \quad (24)$$

and it is *exact*. The proof is three substitutions: the compressed-product display gives $a \cdot b = \text{Co}_{A,C}(ab)$; the imported identities at $n = 1$ give $\text{Co}_{A,C}(ab)^\dagger = \text{Co}_{C,A}((ab)^\dagger)$; the involution axiom gives $(ab)^\dagger = b^\dagger a^\dagger$; and a second use of the display, legitimate because the reversed corner memberships were just established, gives $\text{Co}_{C,A}(b^\dagger a^\dagger) = b^\dagger \cdot a^\dagger$. No associativity estimate and no approximation enter. The tree records a correction here worth one sentence: an earlier audit node concluded that (24) was *not* derivable, because the formula $a \cdot b = \text{Co}_{A,C}(ab)$ had not been registered as an allowed input in the workspace; after the compressed-product display was registered, that negative conclusion — and the associated demand for further definition provisioning — was explicitly withdrawn.

Level one. Fix $z \in \mathcal{S}_{P,R}$ and put $T = \text{Ha}_{P,R}^Q(z)$. The defining relation of `def-ha-map` reads, for all $x \in \mathcal{S}_{R,Q}$ and $y \in \mathcal{S}_{P,Q}$,

$$(y^\dagger \cdot z) \cdot x + y^\dagger \cdot (z \cdot x) = 2 \langle y, Tx \rangle \tilde{Q}. \quad (25)$$

Apply † to (25). On the left, (24) reverses each compressed product, turning the two summands into $x^\dagger \cdot (z^\dagger \cdot y)$ and $(x^\dagger \cdot z^\dagger) \cdot y$. On the right, $\tilde{Q}$ is Hermitian and the scalar is conjugated,

and conjugate symmetry together with the definition of the Hilbert-space adjoint gives $\overline{\langle y, Tx \rangle} = \langle Tx, y \rangle = \langle x, T^\dagger y \rangle$. Thus for all x, y ,

$$(x^\dagger \cdot z^\dagger) \cdot y + x^\dagger \cdot (z^\dagger \cdot y) = 2 \langle x, T^\dagger y \rangle \tilde{Q}.$$

This is precisely the defining relation of **def-ha-map** for the pair (R, P) , the symbol $z^\dagger$, the input y and the test vector x ; and that relation determines its solution uniquely. Hence $\text{Ha}_{R,P}^Q(z^\dagger)y = T^\dagger y$ for every y , i.e. $\text{Ha}_{R,P}^Q(z^\dagger) = \text{Ha}_{P,R}^Q(z)^\dagger$.

Amplification. Write $Z = [Z_{ij}]$. By the amplification convention of the contract, $(\text{Ha}_{P,R}^Q)_n(Z)$ is the block operator $[\text{Ha}_{P,R}^Q(Z_{ij})]$ from the n -fold Hilbert direct sum of $\mathcal{S}_{R,Q}$ to that of $\mathcal{S}_{P,Q}$; its Hilbert-space adjoint therefore has (i, j) block $\text{Ha}_{P,R}^Q(Z_{ji})^\dagger$. On the other side, $(Z^\dagger)_{ij} = Z_{ji}^\dagger$, which lies in $\mathcal{S}_{R,P}$ by the corner symmetry above, so the (i, j) block of $(\text{Ha}_{R,P}^Q)_n(Z^\dagger)$ is $\text{Ha}_{R,P}^Q(Z_{ji}^\dagger)$. The level-one identity equates the two blockwise, and equality of all blocks is (23). Since n and Z were arbitrary, the lemma follows.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb2-amplified-adjointness: 21 nodes, all validated, root validated, taint clean.

Role in the chain. Imports **lem-compcb-amplified-compression-identities**. Being exact rather than approximate, it is what allows a C^* -identity argument to be run on the amplified Ha map: it is used in **lem-hcb3-diagonal-upper-norm** (§22) to identify $(\text{Ha}_{P,P}^Q)_n(Z)^\dagger$ with $(\text{Ha}_{P,P}^Q)_n(Z^\dagger)$ inside $\|T\|^2 = \|T^\dagger T\|$, and it is a declared dependency of the not-yet-validated lower-modulus, off-diagonal-inverse and canonical-closeness lemmas of the remaining H-CB queue.

12 The uniform compressed-unit action

Lemma 11 (`lem-compcb-compressed-unit-action`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular corner satisfies*

$$\|u_T \cdot A - A\| \leq C_{\text{co}} e \|A\|, \quad \|A \cdot u_R - A\| \leq C_{\text{co}} e \|A\|. \quad (26)$$

Transcription note. As spelled out in the tree, the datum is an extended ε - C^* -algebra $\mathcal{A}$, an $n \geq 1$, δ -projections T, R with $T_n = I_n \otimes T$, $R_n = I_n \otimes R$, and an element $A \in \mathcal{S}_{T_n, R_n}$; the compressed units are $u_{T,n} = \text{Co}_{T_n, T_n}(T_n)$ and $u_{R,n} = \text{Co}_{R_n, R_n}(R_n)$. By `lem-compcb-amplified-compression` these equal $I_n \otimes \text{Co}_T(T)$ and $I_n \otimes \text{Co}_R(R)$, which is the notation u_T, u_R of the contract.

Registry contract (`verbatim`).

Uniform compressed-unit action: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that, for $e = \delta + \varepsilon \leq e_{\text{co}}$, every compatible amplified rectangular corner satisfies $\|u_T \cdot A - A\| \leq C_{\text{co}} e \|A\|$ and $\|A \cdot u_R - A\| \leq C_{\text{co}} e \|A\|$.

Proof (prose account of the af-validated tree). The estimate is assembled in three layers: replace the compressed units by the projections, act with the projections in the *ambient* product, then replace the ambient product by the compressed one.

Primitive bounds. Since $\mathcal{A}$ is extended, $M_n \otimes \mathcal{A}$ is an ε - C^* -algebra and T_n, R_n are δ -projections there (adjoints and products amplify entrywise and $\|I_n \otimes Z\| = \|Z\|$). Unpacking the $O(\cdot)$ terms of `def-delta-projection` and `def-compressed-corner` gives universal $a, k < \infty$ and $e_0 \leq 1$ such that every δ -projection S obeys $\|S\| \leq a\delta$ or $\|S\| \leq 1 + ae$ — so $\|T_n\|, \|R_n\| \leq M := \max(a, 1 + a)$ in either branch — and such that each compression is within ke of both $L \cdot R$ orderings. Idempotence of the amplified compression (`lem-compcb-amplified-compression-identities`) also gives the fixed-point identity $A = \text{Co}_{T_n, R_n}(A)$.

Compressed unit versus projection. For $S \in \{T_n, R_n\}$ and $u_S = \text{Co}_{S, S}(S)$ one has $(L_S R_S)(S) = S(S^2)$, and bilinearity gives $S(S^2) - S = S(S^2 - S) + (S^2 - S)$. With $\|S^2 - S\| \leq \delta$, the multiplication bound, and $\|\text{Co}_{S, S} - L_S R_S\| \leq ke$,

$$\|u_S - S\| \leq ke\|S\| + (1 + \varepsilon)\|S\|\delta + \delta \leq K_u e, \quad K_u := kM + 2M + 1, \quad (27)$$

and hence $\|u_S\| \leq M + K_u$, uniformly in n .

Ambient action of the projections. Put $D = T_n(AR_n)$. The fixed-point identity and the compression comparison give $\|A - D\| \leq ke\|A\|$, and one telescopes

$$T_n A - A = T_n(A - D) + (T_n D - D) + (D - A).$$

The outer terms cost $2Mke\|A\|$ and $ke\|A\|$. For the middle term write $Y = AR_n$, so that $T_n D - D = T_n(T_n Y) - T_n Y$; approximate associativity contributes $\varepsilon M^2 \|Y\|$ and $\|T_n^2 - T_n\| \leq \delta$ contributes $2\delta\|Y\|$, and $\|Y\| \leq 2M\|A\|$. Altogether $\|T_n A - A\| \leq K_T e\|A\|$ with $K_T = 2Mk + 2M(M^2 + 2) + k$; the right-hand estimate $\|AR_n - A\| \leq K_R e\|A\|$ is the mirror argument with $D = (T_n A)R_n$ and the same constant. Substituting the units via (27) — $\|(u_{T,n} - T_n)A\| \leq 2K_u e\|A\|$ and symmetrically — gives the ambient form

$$\|u_{T,n} A - A\| \leq K_a e\|A\|, \quad \|A u_{R,n} - A\| \leq K_a e\|A\|, \quad K_a := \max(K_T, K_R) + 2K_u. \quad (28)$$

From ambient to compressed. The pairs $(u_{T,n}, A)$ and $(A, u_{R,n})$ are compatible amplified rectangular pairs in the exact sense required by `lem-compcb-rectangular-product`, which therefore gives universal C_p with $\|u_{T,n} \cdot A - u_{T,n}A\| \leq C_p e \|u_{T,n}\| \|A\|$ and its mirror. Combining with (28) and $\|u_S\| \leq M + K_u$ yields (26) with

$$C_{\text{co}} = K_a + C_p (M + K_u), \quad e_{\text{co}} = \min(e_0, e_p, e_{\text{cmp}}, e_{\text{id}}, 1),$$

all universal and independent of $n, \mathcal{A}, T, R, A$.

A bookkeeping remark: the later nodes of the export re-derive (27)–(28) in “dependency-closed” form, so that the final assembly rests only on validated premises and not on any node still pending at the time. The mathematics is unchanged; the restructuring is what let the tree close bottom-up.

Status. `proved`; `af`: `validated (T0)`. Workspace `proofs/lem-compcb-compressed-unit-action`: 20 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-iden` and `lem-compcb-rectangular-product`. It supplies the unit axiom needed by `lem-compcb-corner-algebra` (§15) — a corner cannot be an approximate C^* -algebra without an approximate unit — and it is the input that drives the diagonal unit estimate `lem-hcb3-diagonal-unit` (§21).

13 The row-column compressed-product estimate

Lemma 12 (`lem-compcb-row-column-product`). *There are universal $C_{rc} < \infty$ and $e_{rc} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{rc}$, P, Q are δ -projections in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and $X, Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$, one has*

$$\|\text{Co}_Q(Y^\dagger X) - Y^\dagger X\| \leq C_{rc} e \|Y\| \|X\|, \quad (29)$$

where $Y^\dagger X$ is the ambient product of the $1 \times n$ row $Y^\dagger$ with the $n \times 1$ column X . Transcription note. The product $Y^\dagger X$ is a single element of $\mathcal{A}$, not a matrix: the row and column contract. This is exactly why `lem-compcb-rectangular-product` does not apply directly — its compatibility clause requires both factors inside one square amplification.

Registry contract (verbatim).

Row-column compressed-product estimate: there are universal $C_{rc} < \infty$ and $e_{rc} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{rc}$, P, Q are δ -projections in an extended ε - C^* -algebra $\mathcal{A}$, $n \geq 1$, and X, Y are in $M_{\{n,1\}} \otimes \mathcal{S}_{\{P,Q\}}$, one has $\|\text{Co}_Q(Y^\dagger X) - Y^\dagger X\| \leq C_{rc} e \|Y\| \|X\|$, where $Y^\dagger X$ is the ambient product of the 1-by- n row $Y^\dagger$ with the n -by-1 column X .

Proof (prose account of the af-validated tree). Set $C_{rc} = C_{co}$ and $e_{rc} = \min(e_{co}, e_{cmp})$ with the constants of `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`, and fix arbitrary admissible data. The whole proof is a change of ambient size: embed the row/column pair as off-diagonal blocks of one square amplification, apply the validated square estimate there, and read the result off the single occupied entry.

The embedding. Use the decomposition $\mathbb{C}^{n+1} = \mathbb{C}^n \oplus \mathbb{C}$ and let

$$C_X = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}, \quad R_Y = \begin{pmatrix} 0 & 0 \\ Y^\dagger & 0 \end{pmatrix}$$

in $M_{n+1} \otimes \mathcal{A}$. Writing $X = (X_i)$ and $Y = (Y_i)$ with entries in $\mathcal{S}_{P,Q}$, the dagger identity of `def-compressed-corner` gives $Y_i^\dagger \in \mathcal{S}_{Q,P}$, so X has entries in $\mathcal{S}_{P,Q}$ and $Y^\dagger$ has entries in $\mathcal{S}_{Q,P}$ with the indicated orientations. Since $e \leq e_{cmp}$, `lem-compcb-amplified-compression` at level $n+1$ identifies $M_{n+1} \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{I_{n+1} \otimes P, I_{n+1} \otimes Q}$ and the transposed statement, so zero-padding puts C_X and R_Y in exactly those compatible corners. The padding is isometric: in the operator-space matrix norms, insertion by a scalar coordinate isometry and extraction by its adjoint are both contractive, hence equalities; and involution is isometric because $\mathcal{A}$ is self-adjoint. Therefore

$$\|C_X\| = \|X\|, \quad \|R_Y\| = \|Y^\dagger\| = \|Y\|, \quad (30)$$

and (R_Y, C_X) is a compatible *square* amplified rectangular pair — the right projection of the first factor and the left projection of the second are both $I_{n+1} \otimes P$.

One occupied entry. Ordinary block multiplication gives

$$R_Y C_X = \text{diag}(0_{n \times n}, Y^\dagger X),$$

every other block containing a zero factor. On the compressed side, `def-compressed-corner` applied to the triple $(I_{n+1} \otimes Q, I_{n+1} \otimes P, I_{n+1} \otimes Q)$ gives $R_Y \cdot C_X = \text{Co}_{I_{n+1} \otimes Q}(R_Y C_X)$, using the

abbreviation $\text{Co}_{S,S} = \text{Co}_S$; and `lem-compcb-amplified-compression` applied to the pair (Q, Q) at level $n + 1$ says $\text{Co}_{I_{n+1} \otimes Q} = 1_{M_{n+1}} \otimes \text{Co}_Q$, an entrywise map. Applying it to the diagonal block matrix above,

$$R_Y \cdot C_X = \text{diag}(0_{n \times n}, \text{Co}_Q(Y^\dagger X)). \quad (31)$$

Conclusion. Since $e \leq e_{\text{co}}$ and (R_Y, C_X) is compatible, `lem-compcb-rectangular-product` yields $\|R_Y \cdot C_X - R_Y C_X\| \leq C_{\text{co}} e \|R_Y\| \|C_X\|$. By (31) the left-hand difference is the zero-padded element with single entry $\text{Co}_Q(Y^\dagger X) - Y^\dagger X$, whose norm equals the norm of that entry by the same corner isometry as in (30); and (30) converts the right-hand side. This is (29) with $C_{\text{rc}} = C_{\text{co}}$. Both thresholds are positive universal constants, so $e_{\text{rc}} > 0$, and universal generalisation over the arbitrary data completes the proof.

Status. `proved`; `af`: `validated` (T0). Workspace `proofs/lem-compcb-row-column-product`: 20 nodes (18 `validated`, 2 superseded nodes archived), root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. This shard was factored out of the column-Hilbert workspace at a recorded third stall: the blocking challenge observed that the validated rectangular-product contract quantifies only over square amplifications, so the pair $(Y^\dagger, X)$ — a $1 \times n$ row against an $n \times 1$ column — simply does not satisfy its hypotheses, and the gap had to be closed by a statement of its own rather than by an instantiation. It has one registry consumer, `lem-hcb-column-hilbert-squared` (§16), where $Y = X$ and the estimate becomes the bridge between the column inner product and the ambient norm.

14 The uniform compressed associator estimate

Lemma 13 (lem-hcb0-compressed-associator). *There are universal $C_{\text{as}} < \infty$ and $e_{\text{as}} > 0$ such that every H -CB datum with $e \leq e_{\text{as}}$ and all compatible amplified rectangular A, B, C satisfy*

$$\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\| \leq C_{\text{as}} e \|A\| \|B\| \|C\|. \quad (32)$$

Transcription note. *The left-hand side is the compressed associator of **def-compressed-associator**; “compatible amplified rectangular” is the datum fixed in §8, here for a composable triple. Both the compressed product and the ambient product are only approximately associative — the ambient one by the ε -associator axiom of an ε - C^* -algebra — so the content is that the two failures add up to a single $O(e)$.*

Registry contract (verbatim).

Uniform compressed associator: there are universal $C_{\text{as}} < \text{infinity}$ and $e_{\text{as}} > 0$ such that every H -CB datum with $e \leq e_{\text{as}}$ and all compatible amplified rectangular A, B, C satisfy $\|(A \cdot B) \cdot C - A \cdot (B \cdot C)\| \leq C_{\text{as}} * e * \|A\| \|B\| \|C\|$.

Proof (prose account of the af-validated tree). Let $C_{\text{co}}, e_{\text{co}}$ be the universal witnesses of lem-compcb-rectangular-product and set

$$e_{\text{as}} = \min(e_{\text{co}}, 1), \quad C_{\text{as}} = 2C_{\text{co}}(2 + C_{\text{co}}) + 4C_{\text{co}} + 1.$$

A product-norm bound. For a compatible pair, the rectangular estimate plus the ambient bound $\|AB\| \leq (1 + \varepsilon)\|A\|\|B\|$ from the ε - C^* norm axioms give

$$\|A \cdot B\| \leq (1 + \varepsilon + C_{\text{co}}e)\|A\|\|B\|, \quad (33)$$

and likewise for $B \cdot C$. This is what keeps the outer terms of the telescope below from degrading.

The telescope. The compressed associator has the exact five-term expansion

$$\begin{aligned} (A \cdot B) \cdot C - A \cdot (B \cdot C) &= [(A \cdot B) \cdot C - (A \cdot B)C] + [(A \cdot B)C - (AB)C] \\ &\quad + [(AB)C - A(BC)] + [A(BC) - A(B \cdot C)] + [A(B \cdot C) - A \cdot (B \cdot C)], \end{aligned} \quad (34)$$

which routes both associations through the ambient product and isolates the ambient associator in the middle.

Write $N = \|A\|\|B\|\|C\|$. The first and fifth terms are rectangular-product errors for the compatible pairs $(A \cdot B, C)$ and $(A, B \cdot C)$, so by (33) each is at most $C_{\text{co}}e(1 + \varepsilon + C_{\text{co}}e)N$. The second and fourth are the rectangular-product errors for (A, B) and (B, C) multiplied by one ambient factor — bilinearity gives $(A \cdot B)C - (AB)C = (A \cdot B - AB)C$ and $A(BC) - A(B \cdot C) = A(BC - B \cdot C)$ — so each is at most $(1 + \varepsilon)C_{\text{co}}eN$. The third is precisely the ε -associator axiom, $\|(AB)C - A(BC)\| \leq \varepsilon N$.

Summation. Adding and using $0 \leq \varepsilon \leq e \leq 1$, so that $1 + \varepsilon + C_{\text{co}}e \leq 2 + C_{\text{co}}$ and $1 + \varepsilon \leq 2$, gives (32) with the stated C_{as} .

Two pieces of hygiene in the tree are worth recording, since they shape the constant. The summation is written so that the ambient-associator term is bounded directly by eN rather than as $(\varepsilon/e) \cdot eN$, because at the endpoint $e = 0$ one has $\varepsilon = 0$ and every error vanishes — the quotient ε/e is never formed. And the upstream witness is normalised to $C_{\text{co}} := \max\{C_{\text{co}}, 0\}$, which cannot

weaken the imported estimate since $e\|A\|\|B\| \geq 0$, but does make the sign-sensitive arithmetic above valid.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb0-compressed-associator: 20 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product`. It is the base of the H-CB tier: the defining relation of the Ha map (`def-ha-map`) pairs an element against precisely the difference of the two associations $(Y^\dagger \cdot Z) \cdot X$ and $Y^\dagger \cdot (Z \cdot X)$, so (32) is exactly the estimate that turns that relation into a quantitative bound. It feeds `lem-hcb1-column-action` (§19) and `lem-hcb2-product-defect` (§20).

15 The compressed corner as an extended approximate C^* -algebra

Lemma 14 (`lem-compcb-corner-algebra`). *There are universal $C_{ca} < \infty$ and $e_{ca} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ca}$ and P is a nonvanishing δ -projection, the corner $\mathcal{S}_P$ — equipped with the compressed product, the inherited involution and the compressed unit $u_P = \text{Co}_P(P)$ — is an extended $C_{ca}e$ - C^* -algebra. Transcription note. “Extended” is as in `def-extended-epsilon-cstar-algebra`: not merely that $\mathcal{S}_P$ satisfies the approximate C^* axioms, but that every $M_n \otimes \mathcal{S}_P$ does, with the same parameter $C_{ca}e$, independent of n . That uniformity is the whole content.*

Registry contract (`verbatim`).

Uniform compressed-corner algebra: there are universal $C_{ca} < \text{infinity}$ and $e_{ca} > 0$ such that, whenever $e = \delta + \varepsilon \leq e_{ca}$ and P is a nonvanishing δ -projection, $\mathcal{S}_P$ with the compressed product, inherited involution, and compressed unit $u_P = \text{Co}_P(P)$ is an extended $C_{ca}e$ - C^* -algebra.

Proof (prose account of the af-validated tree). The tree is organised as one node per axiom family, over a shared constant ledger, with a preliminary step that moves everything to level n at once.

Constant ledger and amplified setup. Let C_r, e_r come from `lem-compcb-rectangular-product`; C_u, e_u from `lem-compcb-compressed-unit-action`; C_N, e_N from `lem-compcb-compressed-unit-norm`; and e_i from `lem-compcb-amplified-compression-identities`. Put

$$K_p = C_r + 1, \quad K_a = 2C_r(K_p + 3) + 1, \quad C_{ca} = \max\{K_p, K_a, C_u, C_N\}, \quad e_{ca} = \min\{e_r, e_u, e_N, e_i, 1\}.$$

With $P_n = I_n \otimes P$, the tree then *derives inside itself* — from `def-compressed-corner` and the registered θ calculus, rather than importing it — the naturality $\text{Co}_{P_n} = \text{id}_{M_n} \otimes \text{Co}_P$: writing $T_P = L_P R_P + R_P L_P - 1$, multiplication by the block-diagonal P_n acts entrywise so $T_{P_n} = \text{id}_{M_n} \otimes T_P$; powers of $T_{P_n} - x_0 I$ amplify by induction; so every partial sum of the θ series is the amplification of the corresponding level-one partial sum, and at fixed finite n the norm limits agree. Consequently $\mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$, the P_n -compressed product of $[X_{ij}], [Y_{jk}]$ has entries $\text{Co}_P(\sum_j X_{ij} Y_{jk}) = \sum_j X_{ij} \cdot Y_{jk}$ — exactly the matrix amplification of the compressed product — and $\text{Co}_{P_n}(P_n) = I_n \otimes \text{Co}_P(P)$. Also P_n is a δ -projection, so all four imported estimates apply at every level.

Operator space and completeness. `lem-compcb-amplified-compression-identities` gives $\text{Co}_{P_n}^2 = \text{Co}_{P_n}$, so $\mathcal{S}_{P_n} = \text{Im } \text{Co}_{P_n}$ is a closed subspace of the complete space $M_n \otimes \mathcal{A}$ and hence complete in the inherited norm. With the inherited matrix norms $\mathcal{S}_P$ is therefore a complete operator space.

Multiplication and associator. By `lem-compcb-rectangular-product` and the ambient bound $\|AB\| \leq (1+\varepsilon)\|A\|\|B\|$, one gets $\|A \cdot B\| \leq (1+K_p e)\|A\|\|B\|$. For the associator, insert successively $(A \cdot B)C$, $(AB)C$, $A(BC)$ and $A(B \cdot C)$ between the two associations; the two outer rectangular errors cost $C_r e(1+K_p e)N$ each, the two errors obtained by multiplying $A \cdot B - AB$ or $BC - B \cdot C$ by an ambient factor cost $(1+\varepsilon)C_r e N$ each, and the ambient associator costs εN , where $N = \|A\|\|B\|\|C\|$. With $\varepsilon \leq e \leq 1$ the total is at most $K_a e N$.

Involution. If $A \in \mathcal{S}_{P_n}$ then $A = \text{Co}_{P_n}(A)$, so the imported dagger identity gives $A^\dagger = \text{Co}_{P_n}(A^\dagger)$ and the inherited isometric involution preserves the corner. Anti-multiplicativity is exact: $(A \cdot B)^\dagger = \text{Co}_{P_n}((AB)^\dagger) = \text{Co}_{P_n}(B^\dagger A^\dagger) = B^\dagger \cdot A^\dagger$. The unit is fixed, $u_{P_n}^\dagger = \text{Co}_{P_n}(P_n^\dagger) = u_{P_n}$, since $P_n^\dagger = P_n$. Finally the lower C^* -norm bound: the rectangular estimate applied to $A^\dagger$ and A , isometry of the involution, and the ambient lower estimate give

$$\|A^\dagger \cdot A\| \geq \|A^\dagger A\| - C_r e \|A\|^2 \geq (1 - \varepsilon - C_r e) \|A\|^2 \geq (1 - (C_r + 1)e) \|A\|^2.$$

Unit. `lem-compcb-compressed-unit-action` applied to the amplified square corner gives $\|u_{P,n} \cdot A - A\| \leq C_{ue}\|A\|$ and its right-hand mirror at every n . For the unit norm, each P_n is a δ -projection with $\|P_n\| = \|P\|$, so the *nonvanishing* alternative passes from P to P_n ; that is exactly the hypothesis under which `lem-compcb-compressed-unit-norm` gives the two-sided $|\|u_{P,n}\| - 1| \leq C_{Ne}$, uniformly in n . This is where the nonvanishing hypothesis of the contract is used, and why it cannot be dropped.

Collecting the four families under the ledger constants gives the extended $C_{ca}e\text{-}C^*$ structure.

Status. `proved`; `af`: `validated` (T0). Workspace `proofs/lem-compcb-corner-algebra`: 14 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression-identities`, `lem-compcb-rectangular-pro`, `lem-compcb-compressed-unit-action` and `lem-compcb-compressed-unit-norm`. It is the output of the COMP tier: it converts a bundle of separate estimates into a single *structure* that later tiers may use as a black box. Its registry consumers are `lem-compcb-single-compression-transfer` (§17), `lem-hcb3-diagonal-upper-norm` (§22) — where the matrix-uniform product axiom is precisely the input that a level-one statement could not supply — and the not-yet-validated `lem-hcb3-diagonal-lower`.

16 The corrected amplified column-Hilbert estimate

Lemma 15 (`lem-hcb-column-hilbert-squared`). *There are universal $C_{\text{col}} < \infty$ and $e_{\text{col}} > 0$ such that every H-CB datum with $e \leq e_{\text{col}}$, every $n \geq 1$ and every $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$\left| \langle X, X \rangle_n - \|X\|_{n,1}^2 \right| \leq C_{\text{col}} e \|X\|_{n,1}^2. \quad (35)$$

Transcription note. $\langle \cdot, \cdot \rangle_n$ is the amplified column inner product of **def-column-hilbert-corner**, defined by $Y^\dagger \cdot X = \langle Y, X \rangle_n u_Q$ with $u_Q = \text{Co}_Q(Q)$, and $\|\cdot\|_{n,1}$ is the ambient norm on the column space. The word “corrected” in the contract refers to the fact that this is the squared comparison: the unsquared display at the corresponding place in the pinned source is known to be false and is deliberately excluded from **def-column-hilbert-corner**.

Registry contract (`verbatim`).

Corrected amplified column-Hilbert estimate: there are universal $C_{\text{col}} < \text{infinity}$ and $e_{\text{col}} > 0$ such that every H-CB datum with $e \leq e_{\text{col}}$, every $n \geq 1$, and every X in $M_{\{n,1\}} \text{ tensor } S_{\{P,Q\}}$ satisfy $\text{abs}(\langle X, X \rangle_n - \|X\|_{\{n,1\}}^2) \leq C_{\text{col}} * e * \|X\|_{\{n,1\}}^2$.

Proof (**prose account of the af-validated tree**). *Scalarisation.* Fix an admissible datum, n and X , and abbreviate

$$r = \|X\|_{n,1}, \quad h = \langle X, X \rangle_n, \quad a = \|u_Q\|, \quad b = \|X^\dagger X\|.$$

The column space is a Hilbert space for its defined inner product, so $h \geq 0$ is real, and the defining amplified display of **def-column-hilbert-corner** at $Y = X$ reads

$$X^\dagger \cdot X = h u_Q. \quad (36)$$

The proof is then the interaction of three scalar estimates for h, a, b, r .

(i) *The compressed-versus-ambient comparison.* Applying `lem-compcb-row-column-product` with $Y = X$ gives $\|\text{Co}_Q(X^\dagger X) - X^\dagger X\| \leq C_{\text{rc}} e r^2$; and $\text{Co}_Q(X^\dagger X) = X^\dagger \cdot X = h u_Q$ by (36). Absolute homogeneity and the reverse triangle inequality therefore give

$$|ha - b| = \left| \|h u_Q\| - \|X^\dagger X\| \right| \leq \|h u_Q - X^\dagger X\| \leq C_{\text{rc}} e r^2. \quad (37)$$

(ii) *The C^* comparison.* The ε - C^* lower norm axiom at the amplified rectangular level gives $b \geq (1 - \varepsilon)r^2$, and submultiplicativity together with isometry of the involution gives $b \leq (1 + \varepsilon)\|X^\dagger\| r = (1 + \varepsilon)r^2$. Hence

$$|b - r^2| \leq \varepsilon r^2 \leq e r^2. \quad (38)$$

(iii) *The compressed unit is close to a unit.* Applying `lem-compcb-compressed-unit-norm` with $T = Q$ gives $|a - 1| \leq Ce$, and with $Ce \leq \frac{1}{2}$ also $a \geq \frac{1}{2}$. This step needs Q *nonvanishing*, which the datum does not literally state — **def-hcb-datum** says only that Q is a *one-dimensional* δ -projection. The tree first records this as a genuine gap, then closes it by a registered bridge: one-dimensionality means $\dim \mathcal{S}_Q = 1$, so $\mathcal{S}_Q \neq 0$; and $\mathcal{S}_Q = 0$ characterises the first (vanishing) alternative of **def-delta-projection**; hence Q lies in the second alternative, i.e. is nonvanishing.

Synthesis. Enlarge the constant and shrink the threshold so that one pair $C \geq 1$, $e_0 \leq 1/(2C)$ serves all three estimates. Then (37) and (38) give $|ha - r^2| \leq (C + 1)er^2$. The exact scalar identity

$$a(h - r^2) = (ha - r^2) + r^2(1 - a) \quad (39)$$

combined with $|a - 1| \leq Ce$ then yields $a|h - r^2| \leq (2C + 1)er^2$, and dividing by $a \geq \frac{1}{2}$ gives (35) with $C_{\text{col}} = 4C + 2$ and $e_{\text{col}} = e_0$.

The withdrawn route. A substantial part of this tree is a *negative* audit, and it is the reason two of the four dependencies exist. The natural first attempt was to square-pad the column, $X_c = Xe_1^*$ and $X_r = X_c^\dagger$, and apply `lem-compcb-rectangular-product` to the pair (X_r, X_c) . The audit found that under this shard's allowed inputs none of the three needed identifications was available: the corner memberships $M_n \otimes \mathcal{S}_{P,Q} = \mathcal{S}_{P_n, Q_n}$ and $(\mathcal{S}_{P,Q})^\dagger \subseteq \mathcal{S}_{Q,P}$, the compressed-product identity $X_r \cdot X_c = E_{11} \otimes (X^\dagger \cdot X)$, and the matrix-norm identifications $\|X_c\| = r$ and $\|E_{11} \otimes z\| = \|z\|$. Each affirmative claim was withdrawn rather than repaired in place; a further over-claim — that the failure of this one construction proved an *exhaustive* impossibility — was also withdrawn when challenged. The actual repair was structural: register `lem-compcb-row-column-product` and `lem-compcb-entrywise-compression-naturality` as registry dependencies, after which step (i) above applies directly, with no square embedding and no E_{11} identification needed at this level.

Status. proved; af: validated (T0). Workspace `proofs/lem-hcb-column-hilbert-squared`: 50 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-rectangular-product`, `lem-compcb-compressed-unit-norm`, `lem-compcb-entrywise-compression-naturality` and `lem-compcb-row-column-product`. It is the analytic hinge of the H-CB tier: it makes the column-Hilbert norm q_P and the ambient norm interchangeable up to $1 \pm O(e)$, which is what lets estimates proved in one be read in the other. It feeds `lem-hcb1-variational-identity` (§18), `lem-hcb1-column-action` (§19) and the not-yet-validated `lem-hcb4-canonical-closeness`.

17 Single-compression transfer of an extended inclusion

Lemma 16 (`lem-compcb-single-compression-transfer`). *There are universal $C_{\text{co}} < \infty$ and $e_{\text{co}} > 0$ such that restricting an extended α -inclusion to one ideal and following it by one compatible amplified compression produces an extended $C_{\text{co}}(\alpha + \varepsilon)$ -inclusion whenever $\alpha + \varepsilon \leq e_{\text{co}}$.
Transcription note. The tree fixes the datum: $\mathcal{A}$ an extended ε - C^* -algebra, $\mathcal{B}$ an exact unital C^* -algebra, $\mathcal{J} \subseteq \mathcal{B}$ a unital direct-summand ideal with unit q , and $v : \mathcal{B} \rightarrow \mathcal{A}$ an extended α -inclusion (*def-extended-delta-inclusion*). The “compatible single compression” is*

$$P := v(q), \quad T := \text{Co}_P \circ v|_{\mathcal{J}},$$

and throughout $e = \alpha + \varepsilon$.

Registry contract (`verbatim`).

Single-compression transfer: there are universal $C_{\text{co}} < \text{infinity}$ and $e_{\text{co}} > 0$ such that restricting an extended α -inclusion to one ideal and following it by one compatible amplified compression produces an extended $C_{\text{co}}(\alpha + \text{epsilon})$ -inclusion whenever $\alpha + \text{epsilon} \leq e_{\text{co}}$.

Proof (prose account of the af-validated tree). *The projection and the ledger.* Since $q = q^\dagger = q^2$ with $\|q\| = 1$ in the exact C^* -algebra $\mathcal{B}$, the inclusion clauses give $P^\dagger = v(q^\dagger) = P$, $\|P^2 - P\| = \|v(q)v(q) - v(q^2)\| \leq \alpha$ and $1 - \alpha \leq \|P\| \leq 1 + \alpha$; so P is an α -projection and, for small e , a *nonvanishing* one, its norm lying in the second alternative of **def-delta-projection**. That is what licenses **lem-compcb-corner-algebra** to make $\mathcal{S}_P$ an extended $C_{\text{cae}}\text{-}C^*$ -algebra — the target structure. The remaining ledger constants are k_{cmp} from the compression comparison $\|\text{Co}_{P_n}(Z) - (P_n Z)P_n\| \leq k_{\text{cmp}}e\|Z\|$, then $D = 2k_{\text{cmp}} + 5$ and $K_N = D + 1$, together with C_r, e_r from **lem-compcb-rectangular-product**.

The exact clauses. By tensor functoriality $T_n = \text{id}_{M_n} \otimes T = (\text{id}_{M_n} \otimes \text{Co}_P) \circ v_n$, and **lem-compcb-amplified-compression-identities** rewrites this as $T_n = \text{Co}_{P_n} \circ v_n$ with range $\text{Im } \text{Co}_{P_n} = \mathcal{S}_{P_n} = M_n \otimes \mathcal{S}_P$; linearity is inherited from v_n and Co_{P_n} . Two clauses hold *exactly*, with no error term. The star clause of the extended inclusion gives $v_n(x^\dagger) = v_n(x)^\dagger$, and **lem-compcb-amplified-compression-identities** for the square pair (P_n, P_n) gives $\text{Co}_{P_n}(Z^\dagger) = \text{Co}_{P_n}(Z)^\dagger$, so $T_n(x^\dagger) = T_n(x)^\dagger$. And the unit of $M_n \otimes \mathcal{J}$ is $q_n = I_n \otimes q$ with $v_n(q_n) = P_n$, so

$$T_n(q_n) = \text{Co}_{P_n}(P_n) = I_n \otimes \text{Co}_P(P) = I_n \otimes u_P :$$

the unit defect is zero.

Compression closeness. Because q is the unit of the ideal, $q_n x = x q_n = x$ for $x \in M_n \otimes \mathcal{J}$, so with $Z = v_n(x)$ the multiplicative clause of the inclusion gives $\|P_n Z - Z\| \leq \alpha\|x\|$ and $\|Z P_n - Z\| \leq \alpha\|x\|$, while its norm clause gives $\|P_n\| \leq 1 + \alpha$ and $\|Z\| \leq (1 + \alpha)\|x\|$. Combining with the compression comparison and $e \leq \frac{1}{2}$,

$$\|T_n(x) - v_n(x)\| \leq \|\text{Co}_{P_n}(Z) - (P_n Z)P_n\| + \|(P_n Z)P_n - Z\| \leq D e \|x\|, \quad (40)$$

uniformly in n ; the second term is bounded by $[(1 + \varepsilon)(1 + \alpha) + 1]\alpha\|x\| \leq 5e\|x\|$.

Norm distortion. The inclusion gives $(1 - \alpha)\|x\| \leq \|v_n(x)\| \leq (1 + \alpha)\|x\|$; with (40) the triangle and reverse-triangle inequalities give

$$(1 - K_N e)\|x\| \leq \|T_n(x)\| \leq (1 + K_N e)\|x\|, \quad K_N = D + 1. \quad (41)$$

Multiplicativity. Insert $v_n(xy)$, $v_n(x)v_n(y)$ and $T_n(x)T_n(y)$ between $T_n(xy)$ and $T_n(x) \cdot T_n(y)$. The first term is $\leq De\|x\|\|y\|$ by (40), using that $M_n \otimes \mathcal{J}$ is an exact C^* -algebra — so its norm is genuinely submultiplicative, $\|xy\| \leq \|x\|\|y\|$. The second is $\leq \alpha\|x\|\|y\|$ by the multiplicative clause. The third, after inserting $T_n(x)v_n(y)$, is $\leq 2D(K_N + 3)e\|x\|\|y\|$ by (40), (41) and the ε -product bound. The fourth needs a naturality observation: for $A, B \in M_n \otimes \mathcal{S}_P$ the amplified compressed product has entries $\sum_k \text{Co}_P(a_{ik}b_{kj}) = \text{Co}_P(\sum_k a_{ik}b_{kj})$, i.e. it equals $\text{Co}_{P_n}(AB)$ — so it is the rectangular compressed product for the compatible triple (P_n, P_n, P_n) , not merely an element of the same space, and **lem-compcb-rectangular-product** applies, giving $\leq C_r(1 + K_N)^2 e\|x\|\|y\|$. Summing, $K_M := D + 1 + 2D(K_N + 3) + C_r(1 + K_N)^2$ works.

Assembly. With $C_{\text{co}} = \max\{1, C_{\text{ca}}, K_N, K_M\}$ and e_{co} the positive minimum of all thresholds used, **def-extended-delta-inclusion** makes each T_n a $C_{\text{co}}e$ -inclusion; since n was arbitrary and every constant is universal, T is an extended $C_{\text{co}}(\alpha + \varepsilon)$ -inclusion, with no dependence on the amplification level or the ideal's dimension.

Status. proved; af: validated (T0). Workspace proofs/lem-compcb-single-compression-transfer: 33 nodes (32 validated, 1 superseded node archived), root validated, taint clean.

Role in the chain. Imports **lem-compcb-amplified-compression**, **lem-compcb-amplified-compression-iden**, **lem-compcb-amplified-almost-containment**, **lem-compcb-corner-algebra** and **lem-compcb-rectangular-product**; it is the only consumer of the almost-containment lemma. Unlike the rest of the COMP tier it is not used by the H-CB tier: it is the transfer primitive aimed at the still-open EXT and Stage-1 assembly, where inclusions have to be pushed through compressions repeatedly. It has no validated registry consumers yet.

18 The amplified Ha variational identity

Lemma 17 (`lem-hcb1-variational-identity`). *There is a universal $e_{\text{var}} > 0$ such that every H-CB datum with $e \leq e_{\text{var}}$, every $n \geq 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$, $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ and $Y \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ satisfy*

$$2 \langle Y, (\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X \rangle_n \text{Co}_Q(Q) = (Y^\dagger \cdot Z) \cdot X - Y^\dagger \cdot (Z \cdot X). \quad (42)$$

*Transcription note. The identity is exact; the threshold only guarantees that the objects involved are defined. Its content is that the “Ha defect” — the difference between letting $\text{Ha}_{P,R}^Q(Z)$ act on a column and simply multiplying by Z — is, after pairing, literally the compressed associator of **def-compressed-associator** for the triple $(Y^\dagger, Z, X)$. Contract amendment. The codified contract originally wrote the normalising factor as u_Q ; the ratified **def-ha-map** identity and the column-Hilbert displays write it as $\tilde{Q} = \text{Co}_Q(Q)$. The contract now spells out $\text{Co}_Q(Q)$. These are definitionally the same element — the compressed unit is $u_Q = \text{Co}_Q(Q)$ — so this was pure notation typing, neither a strengthening nor a weakening.*

Registry contract (verbatim).

Amplified Ha variational identity: there is a universal $e_{\text{var}} > 0$ such that every H-CB datum with $e \leq e_{\text{var}}$, every $n \geq 1$, Z in M_n tensor $\mathcal{S}_{\{P,R\}}$, X in $M_{\{n,1\}}$ tensor $\mathcal{S}_{\{R,Q\}}$, and Y in $M_{\{n,1\}}$ tensor $\mathcal{S}_{\{P,Q\}}$ satisfy $2 \langle Y, (\text{Ha}^Q_{P,R})_n(Z)X - Z \cdot X \rangle_n \text{Co}_Q(Q) = (Y^\dagger \cdot Z) \cdot X - Y^\dagger \cdot (Z \cdot X)$.

Proof (prose account of the af-validated tree). *Threshold.* Take $e_{\text{var}} = \min(e_{\text{cmp}}, e_{\text{col}})$, the thresholds of `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-squared`, and fix an arbitrary datum and arbitrary n, Z, X, Y .

Entrywise dictionary. Write $Z = (Z_{ij})_{1 \leq i,j \leq n}$, $X = (X_j)_j$, $Y = (Y_i)_i$. Matrix amplification, the matrix compressed product and the amplified sesquilinear form unfold as

$$\begin{aligned} ((\text{Ha}_{P,R}^Q)_n(Z)X)_i &= \sum_j \text{Ha}_{P,R}^Q(Z_{ij})(X_j), & (Z \cdot X)_i &= \sum_j Z_{ij} \cdot X_j, \\ (Y^\dagger \cdot Z)_j &= \sum_i Y_i^\dagger \cdot Z_{ij}, & \langle Y, W \rangle_n &= \sum_i \langle Y_i, W_i \rangle. \end{aligned}$$

Summing the level-one relation. For each pair (i, j) , the defining relation of **def-ha-map**, applied to $Z_{ij} \in \mathcal{S}_{P,R}$, $X_j \in \mathcal{S}_{R,Q}$ and $Y_i \in \mathcal{S}_{P,Q}$, gives

$$2 \langle Y_i, \text{Ha}_{P,R}^Q(Z_{ij})(X_j) \rangle \text{Co}_Q(Q) = (Y_i^\dagger \cdot Z_{ij}) \cdot X_j + Y_i^\dagger \cdot (Z_{ij} \cdot X_j).$$

Summing over i, j and using the dictionary together with sesquilinearity,

$$2 \langle Y, (\text{Ha}_{P,R}^Q)_n(Z)X \rangle_n \text{Co}_Q(Q) = \sum_{i,j} \left[(Y_i^\dagger \cdot Z_{ij}) \cdot X_j + Y_i^\dagger \cdot (Z_{ij} \cdot X_j) \right]. \quad (43)$$

The column display. The registered amplified column-Hilbert display $Y^\dagger \cdot W = \langle Y, W \rangle_n \text{Co}_Q(Q)$, applied to $W = Z \cdot X$ and expanded entrywise, gives

$$\langle Y, Z \cdot X \rangle_n \text{Co}_Q(Q) = Y^\dagger \cdot (Z \cdot X) = \sum_{i,j} Y_i^\dagger \cdot (Z_{ij} \cdot X_j). \quad (44)$$

Cancellation. Associating only as displayed — this is the point where the two sides differ — ordinary matrix multiplication gives $(Y^\dagger \cdot Z) \cdot X = \sum_{i,j} (Y_i^\dagger \cdot Z_{ij}) \cdot X_j$, whereas $Y^\dagger \cdot (Z \cdot X) = \sum_{i,j} Y_i^\dagger \cdot (Z_{ij} \cdot X_j)$. Subtracting twice (44) from (43) and using sesquilinearity in the second variable, the doubled $Y_i^\dagger \cdot (Z_{ij} \cdot X_j)$ sum cancels once and reverses sign, leaving exactly (42). Since the datum and n, Z, X, Y were arbitrary and e_{var} is universal and positive, the lemma follows.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb1-variational-identity: 8 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-compcb-amplified-compression-identities` and `lem-hcb-column-hilbert-`

It is the conversion device of the H-CB tier: on its own it proves no estimate, but it turns every bound on the compressed associator into a bound on the Ha defect. Its single consumer is `lem-hcb1-column-action` (§19), where it is combined with `lem-hcb0-compressed-associator` and the nondegeneracy of u_Q .

19 The uniform Ha column action

Lemma 18 (`lem-hcb1-column-action`). *There are universal $C_{\text{act}} < \infty$ and $e_{\text{act}} > 0$ such that every H-CB datum with $e \leq e_{\text{act}}$, every $n \geq 1$, every $Z \in M_n \otimes \mathcal{S}_{P,R}$ and every $X \in M_{n,1} \otimes \mathcal{S}_{R,Q}$ satisfy*

$$q_P((\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X) \leq C_{\text{act}} e \|Z\| q_R(X). \quad (45)$$

Transcription note. $q_P(U) = \langle U, U \rangle_n^{1/2}$ is the column-Hilbert norm of `def-column-hilbert-corner` on $M_{n,1} \otimes \mathcal{S}_{P,Q}$, and likewise q_R ; $\|Z\|$ is the ambient norm.

Registry contract (`verbatim`).

Uniform Ha column action: there are universal $C_{\text{act}} < \text{infinity}$ and $e_{\text{act}} > 0$ such that every H-CB datum with $e \leq e_{\text{act}}$, every $n \geq 1$, Z in M_n tensor $\mathcal{S}_{\{P,R\}}$, and X in $M_{\{n,1\}}$ tensor $\mathcal{S}_{\{R,Q\}}$ satisfy $q_P((\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X) \leq C_{\text{act}} * e * \|Z\| * q_R(X)$.

Proof (prose account of the af-validated tree). Throughout put $D := (\text{Ha}_{P,R}^Q)_n(Z)X - Z \cdot X$, the object to be bounded.

Nondegeneracy of the compressed unit. Everything is divided by $\|u_Q\|$ at the end, so a lower bound is needed. Two ingredients combine. First, $\|u_Q - Q\| \leq K_1 e$. This is proved *inside the associative operator algebra* $B(\mathcal{A})$, not in the ambient ε - C^* -algebra, which is only approximately associative — a distinction the tree makes explicitly. With $L = L_Q$, $R = R_Q$ and $A = LR + RL - I$ one has $\text{Co}_Q = \theta(A)$; the δ -projection dichotomy bounds $\|Q\|$, hence $\|L\|$, $\|R\|$, and the ε -associator and δ -projection bounds give $\|LR - RL\| = O(e)$, $\|L^2 - L\| = O(e)$, $\|R^2 - R\| = O(e)$; from these, $T = \frac{1}{2}(LR + RL)$ satisfies $\|T^2 - T\| \leq C_T e$ and so $\|A^2 - I\| = 4\|T^2 - T\| \leq 4C_T e$. Writing $r = Q^2 - Q$ and keeping the displayed parentheses, $(A - I)Q = 2r + Qr + rQ$, so $\|(A - I)Q\| \leq C_Q e$. Finally, below the threshold making $\|A^2 - I\| < \frac{1}{2}$, the series $B = (A^2)^{-1/2}$ converges and factors as $B - I = (A^2 - I)K(A^2)$; since A commutes with B and $K(A^2)$, one gets the clean factorisation $\theta(A) - I = (A - I)H(A)$ with $\|H(A)\| \leq C_H$, whence $\|u_Q - Q\| = \|(\theta(A) - I)Q\| \leq C_H C_Q e$. Second, $\|Q\| \geq 1 - K_0 e$, which holds *provided Q is nonvanishing*. The tree runs this branch strictly conditionally — it refuses to infer nonvanishing from one-dimensionality — and discharges the antecedent only later, from the registered bridge $\dim \mathcal{S}_Q = 1 \Rightarrow \mathcal{S}_Q \neq 0 \Rightarrow Q$ nonvanishing. Combining the two by the reverse triangle inequality and shrinking the threshold gives

$$\|u_Q\| \geq c_u := \frac{1}{2}. \quad (46)$$

Column-norm comparison. By `lem-hcb-column-hilbert-squared`, $|q_T(U)^2 - \|U\|_{n,1}^2| \leq C_{\text{col}} e \|U\|_{n,1}^2$ for $T = P$ and $T = R$; with $C_{\text{col}} \leq \frac{1}{2}$ all terms are nonnegative and taking square roots gives $(1 - C_{\text{col}} e)^{1/2} \|U\|_{n,1} \leq q_T(U) \leq (1 + C_{\text{col}} e)^{1/2} \|U\|_{n,1}$, in particular $\|U\|_{n,1} \leq \sqrt{2} q_T(U)$.

Defect pairing. By `lem-hcb1-variational-identity`, $2 \langle Y, D \rangle_n u_Q = (Y^\dagger \cdot Z) \cdot X - Y^\dagger \cdot (Z \cdot X)$ for every compatible column Y . The right-hand side is precisely the compressed associator of $A = Y^\dagger$, $B = Z$, $C = X$ (`def-compressed-associator`), so `lem-hcb0-compressed-associator` bounds it by $C_{\text{as}} e \|Y^\dagger\| \|Z\| \|X\|_{n,1}$, and the involution is isometric, so $\|Y^\dagger\| = \|Y\|_{n,1}$. Taking norms on the left and using absolute homogeneity,

$$2 |\langle Y, D \rangle_n| \|u_Q\| \leq C_{\text{as}} e \|Y\|_{n,1} \|Z\| \|X\|_{n,1}. \quad (47)$$

Assembly by self-pairing. Insert (46) and the two norm comparisons into (47): for every compatible Y ,

$$|\langle Y, D \rangle_n| \leq 2C_{\text{as}} e q_P(Y) \|Z\| q_R(X).$$

Now the finishing move avoids any duality argument: since both $(\text{Ha}_{P,R}^Q)_n(Z)X$ and $Z \cdot X$ lie in $M_{n,1} \otimes \mathcal{S}_{P,Q}$, so does D , and $Y = D$ is an admissible test vector. Substituting it gives $q_P(D)^2 = \langle D, D \rangle_n \leq 2C_{\text{as}} e \|Z\| q_R(X) q_P(D)$; if $q_P(D) = 0$ the conclusion is immediate, and otherwise one divides. Hence (45) with $C_{\text{act}} = 2C_{\text{as}}$.

Two recorded refinements. The tree first proved the terminal cancellation only *conditionally*, explicitly declining to divide by an unavailable c_u and carrying a node that says so; the unconditional root appears only after the nonvanishing bridge is registered, and the earlier negative conclusion is marked stale rather than deleted. Separately, the threshold is set as $e_{\text{act}} = t/2$ for $t = \min(e_u, e_{\text{col}}, e_{\text{var}}, e_{\text{as}}, 1/(2 \max(C_{\text{col}}, 1)))$ so that the strict inequalities required upstream hold even at the endpoint, and the degenerate case $C_{\text{col}} = 0$ is split off so that the reciprocal $1/(2C_{\text{col}})$ is never formed.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb1-column-action: 38 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-hcb-column-hilbert-squared`, `lem-hcb0-compressed-associator` and `lem-hcb1-variational-identity`. This is the first genuinely quantitative statement about the Ha map: it says that $(\text{Ha}_{P,R}^Q)_n(Z)$ is, to within $O(e)$, left multiplication by Z . Everything later in the tier is built on it — `lem-hcb2-product-defect` (§20), `lem-hcb3-diagonal-unit` (§21) — and it is a declared dependency of the not-yet-validated `lem-hcb4-canonical-closeness`.

20 The uniform amplified Ha product defect

Lemma 19 (`lem-hcb2-product-defect`). *There are universal $C_{\text{prod}} < \infty$ and $e_{\text{prod}} > 0$ such that every H-CB datum with $e \leq e_{\text{prod}}$, every $n \geq 1$, every $Z \in M_n \otimes \mathcal{S}_{P,S}$ and every $W \in M_n \otimes \mathcal{S}_{S,R}$ satisfy*

$$\|(\text{Ha}_{P,R}^Q)_n(Z \cdot W) - (\text{Ha}_{P,S}^Q)_n(Z) (\text{Ha}_{S,R}^Q)_n(W)\| \leq C_{\text{prod}} e \|Z\| \|W\|. \quad (48)$$

Transcription note. *On the left the juxtaposition of the two amplified Ha maps is composition of operators between amplified column spaces, whereas $Z \cdot W$ is the compressed product in the ambient corner; the norm is the operator norm of **def-ha-map**. So the statement is that $\text{Ha}_{\cdot}^Q$ is multiplicative up to $O(e)$.*

Registry contract (verbatim).

Uniform amplified Ha product defect: there are universal $C_{\text{prod}} < \text{infinity}$ and $e_{\text{prod}} > 0$ such that every H-CB datum with $e \leq e_{\text{prod}}$, every $n \geq 1$, Z in M_n tensor $S_{\{P,S\}}$, and W in M_n tensor $S_{\{S,R\}}$ satisfy $\|(\text{Ha}^Q_{P,R})_n(Z \cdot W) - (\text{Ha}^Q_{P,S})_n(Z) (\text{Ha}^Q_{S,R})_n(W)\| \leq C_{\text{prod}} * e * \|Z\| \|W\|$.

Proof (prose account of the af-validated tree). *Calibration.* Three uniform facts are set up first. The whole-column identity $X^\dagger \cdot X = q_T(X)^2 u_Q$ of **def-column-hilbert-corner**, combined with **lem-compcb-rectangular-product** and the ε - C^* norm axioms, gives $|\|X^\dagger \cdot X\| - \|X\|^2| \leq (C_{\text{co}} + 1)e\|X\|^2$; since Q is one-dimensional it is nonvanishing by the registered bridge, so u_Q is the unit of an $O(e)$ - C^* corner and its norm is within $O(e)$ of 1. Shrinking the threshold, this calibrates the column norm against the ambient norm,

$$\frac{1}{2}\|X\| \leq q_T(X) \leq 2\|X\|. \quad (49)$$

Next, the rectangular estimate and $\|AB\| \leq (1 + \varepsilon)\|A\|\|B\|$ give $\|A \cdot B\| \leq K_0\|A\|\|B\|$ with $K_0 = 2 + C_{\text{co}}$; and for a square-by-column pair, (49) converts this into the multiplier bound $q_P(A \cdot X) \leq K_{\text{mul}}\|A\|q_R(X)$ with $K_{\text{mul}} = 4K_0$. Finally, **lem-hcb1-column-action** together with the multiplier bound gives a uniform operator bound for the Ha map itself,

$$\|(\text{Ha}_{A,B}^Q)_n(U)\| \leq K_H\|U\|, \quad K_H := K_{\text{mul}} + C_{\text{act}}. \quad (50)$$

The telescope. Abbreviate $H_{A,B}(U) := (\text{Ha}_{A,B}^Q)_n(U)$. For every compatible column X the following identity is exact, obtained by adding and subtracting compressed multiplications:

$$\begin{aligned} [H_{P,R}(Z \cdot W) - H_{P,S}(Z)H_{S,R}(W)]X &= [H_{P,R}(Z \cdot W)X - (Z \cdot W) \cdot X] \\ &\quad + [(Z \cdot W) \cdot X - Z \cdot (W \cdot X)] \\ &\quad + Z \cdot [W \cdot X - H_{S,R}(W)X] \\ &\quad + [Z \cdot H_{S,R}(W)X - H_{P,S}(Z)H_{S,R}(W)X]. \end{aligned} \quad (51)$$

Each bracket is one of the two estimates already available. The first is a column-action defect for the element $Z \cdot W$, so **lem-hcb1-column-action** and $\|Z \cdot W\| \leq K_0\|Z\|\|W\|$ bound it by $C_{\text{act}}K_0e\|Z\|\|W\|q_R(X)$. The second is the compressed associator, bounded by **lem-hcb0-compressed-associator** in the ambient norm by $C_{\text{as}}e\|Z\|\|W\|\|X\|$; converting both ends through (49) costs two factors of 2, giving $4C_{\text{as}}e\|Z\|\|W\|q_R(X)$. The third is the column-action defect for W , pushed through

the square-by-column multiplier bound for Z , giving $K_{\text{mul}}C_{\text{act}}e\|Z\|\|W\|q_R(X)$. The fourth is the column-action defect for Z evaluated on the column $H_{S,R}(W)X$, whose column norm is controlled by the operator bound (50), giving $C_{\text{act}}K_H e\|Z\|\|W\|q_R(X)$.

Conclusion. Applying the q_P triangle inequality to (51) and adding the four bounds gives the pointwise estimate with

$$C_* := C_{\text{act}}K_0 + 4C_{\text{as}} + C_{\text{act}}K_{\text{mul}} + C_{\text{act}}K_H.$$

Taking the supremum over columns with $q_R(X) = 1$ converts it into the operator-norm statement (48), with $C_{\text{prod}} = C_*$ and $e_{\text{prod}} = \min\{e_{\text{cmp}}, e_{\text{co}}, e_{\text{as}}, e_{\text{act}}, e_q, 1\}$; `lem-compcb-amplified-compression` is what guarantees that all the corners and compressed products appearing at level n are the compatible amplified objects the imported estimates quantify over.

Status. `proved`; `af`: `validated (T0)`. Workspace `proofs/lem-hcb2-product-defect`: 12 nodes, all `validated`, root `validated`, taint clean.

Role in the chain. Imports `lem-hcb0-compressed-associator`, `lem-hcb1-column-action`, `lem-compcb-rectangular-product` and `lem-compcb-amplified-compression`. Together with the exact adjointness of §11 it makes $\text{Ha}_{\cdot}^Q$ an approximate $*$ -homomorphism at every amplification level, which is exactly the input the norm estimates of the next tier consume: it feeds `lem-hcb3-diagonal-upper-norm` (§22) and the not-yet-validated `lem-hcb3-diagonal-lower-modulus` and `lem-hcb3-offdiagonal-inverse`.

21 The uniform diagonal Ha unit estimate

Lemma 20 (`lem-hcb3-diagonal-unit`). *There are universal $C_{\text{unit}} < \infty$ and $e_{\text{unit}} > 0$ such that every H-CB datum with $e \leq e_{\text{unit}}$ and every $n \geq 1$ satisfy*

$$\|(\text{Ha}_{P,P}^Q)_n(I_n \otimes u_P) - I\| \leq C_{\text{unit}} e. \quad (52)$$

Transcription note. $u_P = \text{Co}_P(P)$ is the compressed unit of the corner $\mathcal{S}_P$, $I_n \otimes u_P$ its diagonal amplification, and I the identity operator on the amplified column space $M_{n,1} \otimes \mathcal{S}_{P,Q}$; the norm is the operator norm.

Registry contract (verbatim).

Uniform diagonal Ha unit estimate: there are universal $C_{\text{unit}} < \text{infinity}$ and $e_{\text{unit}} > 0$ such that every H-CB datum with $e \leq e_{\text{unit}}$ and every $n \geq 1$ satisfy $\|(\text{Ha}^Q_{P,P})_n(I_n \text{ tensor } u_P) - I\| \leq C_{\text{unit}} * e$.

Proof (prose account of the af-validated tree). Write $U = I_n \otimes u_P$. The estimate is pointwise on columns, then a supremum.

A uniform bound on the amplified unit. The obvious route — bound $\|u_P\|$ at level one and appeal to isometry of the diagonal amplification — is *not* taken, and the tree says why: no registered input supplies that isometry. Instead $\|U\|$ is bounded directly at level n by a C^* -style argument. Put $t = \|U\|$. The ε - C^* norm axioms at level n give $\|U^\dagger U\| \geq (1 - \varepsilon)t^2 \geq (1 - e)t^2$ and $\|U^\dagger\| = t$; the compressed-product display in the amplified P, P corner gives $\|U^\dagger U - U^\dagger \cdot U\| \leq C_{\text{pr}} e t^2$; and `lem-compcb-compressed-unit-action`, in its right-unit form with $A = U^\dagger$, gives $\|U^\dagger \cdot U - U^\dagger\| \leq C_{\text{co}} e t$. Chaining,

$$(1 - e)t^2 \leq \|U^\dagger U\| \leq C_{\text{pr}} e t^2 + (1 + C_{\text{co}} e)t.$$

If $t = 0$ there is nothing to prove; otherwise divide by t to get $[1 - (1 + C_{\text{pr}})e]t \leq 1 + C_{\text{co}} e$, and choosing $e_u \leq 1/[2(1 + C_{\text{pr}})]$ makes the bracket at least $\frac{1}{2}$, so

$$\|U\| \leq K_u := 2(1 + C_{\text{co}} e_u), \quad (53)$$

uniformly in n .

Two defects. For a column $X \in M_{n,1} \otimes \mathcal{S}_{P,Q}$ split

$$((\text{Ha}_{P,P}^Q)_n(U) - I)X = ((\text{Ha}_{P,P}^Q)_n(U)X - U \cdot X) + (U \cdot X - X). \quad (54)$$

The first bracket is a column-action defect: `lem-hcb1-column-action` with $R = P$ and $Z = U$ gives $q_P((\text{Ha}_{P,P}^Q)_n(U)X - U \cdot X) \leq C_{\text{act}} e \|U\| q_P(X)$, which (53) turns into $C_{\text{act}} K_u e q_P(X)$.

The second bracket says that the amplified compressed unit is an approximate identity on columns, and it is assembled coordinatewise. Multiplication by the block-diagonal $U = I_n \otimes u_P$ acts entrywise, so writing $X = (X_1, \dots, X_n)^\top$ and applying `lem-compcb-compressed-unit-action` in each level-one P, Q corner gives $\|u_P \cdot X_j - X_j\| \leq C_{\text{co}} e \|X_j\|$ for every j . The level-one norm/inner-product comparison registered with `def-ha-map` and the column-Hilbert displays supplies a universal κ with $q_P(Y) \leq \kappa \|Y\|$ and $\|Y\| \leq \kappa q_P(Y)$, so each coordinate estimate becomes $q_P(u_P \cdot X_j - X_j) \leq \kappa^2 C_{\text{co}} e q_P(X_j)$. Finally the coordinate-sum display $q_P((Y_j)_j)^2 = \sum_j q_P(Y_j)^2$ lets the squared coordinate estimates be summed, giving $q_P(U \cdot X - X) \leq \kappa^2 C_{\text{co}} e q_P(X)$ uniformly

in n ; the existential constant C_{co} is then simply replaced by the larger universal $\kappa^2 C_{\text{co}}$ and the notation retained.

Conclusion. Setting $e_0 = \frac{1}{2} \min(e_u, e_{\text{act}}, e_{\text{co}})$ and applying the q_P triangle inequality to (54),

$$q_P\left(\left((\text{Ha}_{P,P}^Q)_n(U) - I\right)X\right) \leq (C_{\text{act}}K_u + C_{\text{co}})e_{q_P(X)},$$

and passing to the operator norm gives (52) with $C_{\text{unit}} = C_{\text{act}}K_u + C_{\text{co}}$ and $e_{\text{unit}} = e_0$.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb3-diagonal-unit: 18 nodes, all validated, root validated, taint clean.

Role in the chain. Imports lem-hcb1-column-action and lem-compcb-compressed-unit-action.

It is one of the estimates the parent contract conj-hcb collects — the clause asserting that the amplified Ha map is approximately unital — and it has no validated registry consumers of its own. conj-hcb is proved-mod-audit and af: none: it is not rigorous here, and neither it nor this lemma bears on the status of op-classical, which remains open.

22 The uniform diagonal Ha upper norm

Lemma 21 (`lem-hcb3-diagonal-upper-norm`). *There are universal $C_{\text{up}} < \infty$ and $e_{\text{up}} > 0$ such that every H -CB datum with $e \leq e_{\text{up}}$, every $n \geq 1$ and every $Z \in M_n \otimes \mathcal{S}_P$ satisfy*

$$\|(\text{Ha}_{P,P}^Q)_n(Z)\| \leq (1 + C_{\text{up}} e) \|Z\|. \quad (55)$$

Transcription note. *The left norm is the operator norm on maps of the amplified column space; $\|Z\|$ is the ambient norm of Z in $M_n \otimes \mathcal{S}_P$. The estimate is uniform in n , which is the whole difficulty: the level-one statement is a routine consequence of the corner structure, the matrix-uniform one is not.*

Registry contract (verbatim).

Uniform diagonal Ha upper norm: there are universal $C_{\text{up}} < \text{infinity}$ and $e_{\text{up}} > 0$ such that every H -CB datum with $e \leq e_{\text{up}}$, every $n \geq 1$, and every Z in M_n tensor $\mathcal{S}_P$ satisfy $\|(\text{Ha}^Q_{P,P})_n(Z)\| \leq (1 + C_{\text{up}} * e) \|Z\|$.

Proof (prose account of the af-validated tree). Write $h_n = (\text{Ha}_{P,P}^Q)_n$ and $b_n = \|h_n\|$ for its operator norm; the goal is $b_n \leq 1 + C_{\text{up}} e$ for every n .

The matrix-uniform square estimate. The first ingredient is

$$\|Z^\dagger \cdot Z\| \leq (1 + \kappa e) \|Z\|^2, \quad Z \in M_n \otimes \mathcal{S}_P, \quad (56)$$

with $\kappa = C_{\text{ca}}$ and threshold $e_{\text{sq}} = e_{\text{ca}}$. If $\mathcal{S}_P = 0$ then $M_n \otimes \mathcal{S}_P = 0$ for every n , so $Z = 0$ and (56) is trivial. Otherwise $\mathcal{S}_P \neq 0$ puts P in the nonvanishing branch of the δ -projection dichotomy, and `lem-compcb-corner-algebra` makes $\mathcal{S}_P$ an *extended* $C_{\text{ca}}e$ - C^* -algebra — so, by the meaning of “extended”, every $M_n \otimes \mathcal{S}_P$ is a $C_{\text{ca}}e$ - C^* -algebra with the *same* parameter, independent of n . Its product upper axiom together with isometry of the inherited involution then gives (56).

This is the shard’s instructive correction, and it explains the shape of the dependency list. An earlier audit of the tree found that the permitted inputs at that time — the compressed-product display, the level-one ε - C^* axioms, `def-hcb-datum` (data and notation only), and the two Ha lemmas — gave (56) at $n = 1$ and *nowhere else*: passing to arbitrary n was not an inference from any allowed premise but a genuinely missing amplified-corner theorem. Once `lem-compcb-corner-algebra` became an allowed dependency, the negative audit nodes became obsolete and are marked as such.

A quadratic recurrence for b_n . Finiteness of b_n is established without any finite-dimensionality assumption on $\mathcal{S}_P$. From the defining Ha relation and the level-one bounds — $\|A \cdot B\| \leq K_{\text{dot}} \|A\| \|B\|$, the two-sided comparison $c_q^{-1} \|X\| \leq q(X) \leq c_q \|X\|$, and $\|u_Q\| \geq c_u > 0$ (with Q nonvanishing because it is one-dimensional) — one gets $2|\langle Y, h(Z)X \rangle| \|u_Q\| \leq 2K_{\text{dot}}^2 \|Y\| \|Z\| \|X\|$, hence $|\langle Y, h(Z)X \rangle| \leq K_1 q(Y) \|Z\| q(X)$ with $K_1 = K_{\text{dot}}^2 c_q^2 / c_u$, and Hilbert-space duality in the q norm gives $\|h(Z)\| \leq K_1 \|Z\|$ at level one. For fixed finite n , $h_n(Z) = [h(Z_{ij})]$ on the Hilbert direct sum, so $\|h_n(Z)\| \leq (\sum_{i,j} \|h(Z_{ij})\|^2)^{1/2} \leq n K_1 \|Z\|$, using contractivity of matrix-coordinate compressions. Crude in n , but finite — which is all that is needed to *form* b_n and take a supremum.

Now let $e \leq \min(e_{\text{sq}}, e_{\text{prod}}, e_{\text{adj}})$ and $Z \in E_n := M_n \otimes \mathcal{S}_P$. Put $T = h_n(Z)$. The operator C^* -identity gives $\|T\|^2 = \|T^\dagger T\|$; `lem-hcb2-amplified-adjointness` identifies $T^\dagger = h_n(Z^\dagger)$ — exactly, which is what makes this move legitimate; and `lem-hcb2-product-defect` with all

three corner indices equal to P gives $\|h_n(Z^\dagger)h_n(Z) - h_n(Z^\dagger \cdot Z)\| \leq C_{\text{prod}}e\|Z\|^2$. Chaining with boundedness of h_n and (56),

$$\|h_n(Z)\|^2 \leq b_n\|Z^\dagger \cdot Z\| + C_{\text{prod}}e\|Z\|^2 \leq [(1 + \kappa e)b_n + C_{\text{prod}}e]\|Z\|^2.$$

If $E_n = \{0\}$ then $b_n = 0$ and the recurrence below holds trivially; otherwise the unit sphere of E_n is nonempty, b_n is a finite supremum over it, and $\sup\|h_n(Z)\|^2 = (\sup\|h_n(Z)\|)^2 = b_n^2$, so

$$b_n^2 \leq (1 + \kappa e)b_n + C_{\text{prod}}e. \quad (57)$$

The root argument. It remains to solve (57). For nonnegative b, e, κ, C with $b^2 \leq ab + Ce$ and $a = 1 + \kappa e$, set $x = 1 + (\kappa + C)e$ and $f(t) = t^2 - at - Ce$. Direct expansion gives $f(x) = C(\kappa + C)e^2 \geq 0$, and f is strictly increasing on $[x, \infty)$ because $2x - a = 1 + (\kappa + 2C)e > 0$. The hypothesis is $f(b) \leq 0$; if $b > x$ then monotonicity would give $f(b) > f(x) \geq 0$, a contradiction. Hence $b \leq x$.

Applying this with $b = b_n$ and $C = C_{\text{prod}}$ gives $b_n \leq 1 + (\kappa + C_{\text{prod}})e$ for every n , which is (55) with $C_{\text{up}} = \kappa + C_{\text{prod}}$ and $e_{\text{up}} = \min(e_{\text{sq}}, e_{\text{prod}}, e_{\text{adj}})$; both are universal.

Status. proved; af: validated (T0). Workspace proofs/lem-hcb3-diagonal-upper-norm: 30 nodes, all validated, root validated, taint clean.

Role in the chain. Imports `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect` and `lem-compcb-corner-algebra`. It is the upper half of the norm control that the parent contract `conj-hcb` requires; the matching lower half, `lem-hcb3-diagonal-lower-modulus`, is *not* validated — it is mid-validation at the time of writing — and the inverse and canonical tiers beyond it are af: none. `conj-hcb` itself is `proved-mod-audit`, i.e. open at this repository's rigour ladder, and nothing here bears on the status of `op-classical`.

23 Status and outlook

What is validated here

The preceding sections reproduce twenty-one registry results, each proved with `af: validated:` the signed-stochastic bridge (§2); positive-retract hardening (§3); the COMP tier — `lem-compcb-amplification-r`, `lem-compcb-amplified-compression`, `lem-compcb-amplified-compression-identities`, `lem-compcb-entryw`, `lem-compcb-rectangular-product`, `lem-compcb-amplified-almost-containment`, `lem-compcb-compressed-un`, `lem-compcb-compressed-unit-action`, `lem-compcb-row-column-product`, `lem-compcb-corner-algebra`, `lem-compcb-single-compression-transfer`; and the H-CB tier — `lem-hcb0-compressed-associator`, `lem-hcb-column-hilbert-squared`, `lem-hcb1-variational-identity`, `lem-hcb1-column-action`, `lem-hcb2-amplified-adjointness`, `lem-hcb2-product-defect`, `lem-hcb3-diagonal-unit`, `lem-hcb3-diagonal`. These are the results of the *live* route. The registry's total count of `af: validated` results is larger, because it also carries the validated signed-geometry material listed at the end of this section; that material is off the live route and is not reproduced here.

What is next, and what it is not

The immediate queue continues the H-CB tier bottom-up. In order: `lem-hcb3-diagonal-lower-modulus` (the lower-modulus companion to §22; mid-validation at the time of writing, hence excluded from this document), then `lem-hcb3-diagonal-inverse` and `lem-hcb3-offdiagonal-inverse`, then the canonical tier `lem-hcb4-canonical-gram`, `lem-hcb4-canonical-inverse` and `lem-hcb4-canonical-closene`. Only when all of these are validated can the parent contract `conj-hcb` be attempted; `conj-hcb` is at present `proved-mod-audit` with `af: none` — a hostile-verified paper proof, one rung below rigorous here. Its contract is deliberately *conditional*: an earlier unconditional inverse clause was refuted by an exact $\mathbb{C} \oplus \mathbb{C}$ counterexample and replaced by the verifier's conditional form, which is why the inverse lemmas above carry level-one hypotheses.

Beyond `conj-hcb` lies the EXT tier (`conj-extcb` and the `lem-extcb*` family), and beyond that the assembly. Two items are explicitly *not* in the pipeline and must not be treated as pending detail: the MAIN-CB assembly rows and the ledger rows are **quarantined** pending a focused repair and a fresh hostile review, along with five named gap interfaces (exact-target approximation, the Stage-1 polar contract, the MAIN structure, the ledger domains, and the F2/F3 bridge); and several Stage-1 external topology sources are not yet in `refs/`, which is a provisioning risk, not a proof step.

The honest summary is therefore: the base of one route has been made rigorous, the middle is in progress, and the top is quarantined. `op-classical` is **open**.

Results removed from the previous report

The previous edition of this lab-book documented the signed-geometry route — height collapse, the halo-collapse family, the *(EX)*/factorization and fan-payment chain, the hiddenness witnesses, the slab and capacity primitives, the starvation obstruction and the `hx-` financing engine. That route is not the live one, and those sections have been retired from the paper-track. This is a route pivot, not a retraction: every result below keeps its registry shard, its `af` workspace and its status, and any of them may be re-wired into the paper-track if the route returns to them.

Registry id	Status	Note
obs-height-collapse	proved, validated	af: off the live route; registry record retained
lem-mass-split	proved, validated	af: off the live route; registry record retained
lem-residual-lower	proved, validated	af: off the live route; registry record retained
lem-residual-upper	proved, validated	af: off the live route; registry record retained
lem-halo-collapse	proved, validated	af: off the live route; registry record retained
lem-parametric-halo-collapse	proved, validated	af: off the live route; registry record retained
lem-depth-d-halo-collapse	proved, validated	af: off the live route; registry record retained
lem-genuine-disintegration	proved, validated	af: off the live route; registry record retained
lem-top-concentration	proved, validated	af: off the live route; registry record retained
lem-factorization	proved, validated	af: off the live route; registry record retained
lem-zerosum-triangle	proved, validated	af: off the live route; registry record retained
lem-weighted-min	proved, validated	af: off the live route; registry record retained
lem-fan-payment	proved, validated	af: off the live route; registry record retained
lem-negpart-subadditive	proved, validated	af: off the live route; registry record retained
lem-fan-payment-restricted	proved, validated	af: off the live route; registry record retained
lem-pivot-removing-move	proved, validated	af: off the live route; registry record retained
lem-hiddenness-dual-witness	proved, validated	af: off the live route; registry record retained
lem-always-tight-dual-support	proved, validated	af: off the live route; registry record retained
lem-hiddenness-depth-markov	proved, validated	af: off the live route; registry record retained
lem-row-far-dual-certificate	proved, validated	af: off the live route; registry record retained
lem-top-slab-companion	proved, validated	af: off the live route; registry record retained
lem-cs-low-slab-pincer	proved, validated	af: off the live route; registry record retained
lem-row-zero-capacity	proved, validated	af: off the live route; registry record retained

Registry id	Status	Note
lem-harmonic-affine-bridge	proved, validated	af: off the live route; registry record retained
lem-starvation-completion-obstruction	proved, validated	af: off the live route; registry record retained
lem-hx-transverse-moment-identity	proved, validated	af: off the live route; registry record retained
lem-hx-signed-variation-ledger	proved, validated	af: off the live route; registry record retained
lem-hx-financing-floor	proved, validated	af: off the live route; registry record retained
lem-hx-robust-scalar-starvation	proved, validated	af: off the live route; registry record retained
lem-hx-forced-exterior-coupling	proved, validated	af: off the live route; registry record retained

lem-classical-equiv appeared in the previous edition and is retained here (§2): the bridge is route-independent. The non-rigorous registry rows anchored by the previous edition's status ledger — the inherited `proved-mod-audit` theorems, the `conjecture` rows including `conj-ex` and `conj-kernel`, the `numerical` observations and the `heuristic` arguments — are unaffected by this pivot; they keep their tags in `argument/` and, as stated in §1, are not restated as theorems in this document.